

ISC Class XII Physics

Chapter 3 Electric Resistance and Ohm's Law

77 Numericals — Step-by-Step ISC Board Solutions

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- I. Electric Current and Current Density
- II. Drift Velocity and Mobility
- III. Resistance and Resistivity
- IV. Change in Resistance on Stretching
- V. Effect of Temperature on Resistance
- VI. Combination of Resistances
- VII. Electric Power and Energy
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A note on circuit figures: The original textbook diagrams for the network/circuit questions in this key have been redrawn here as clean, schematic circuit diagrams consistent with the given component values and (where stated) the expected answer. If your textbook's figure looks visually different in layout, the *method* shown — series-parallel reduction, Kirchhoff's laws (KCL + KVL), the Wheatstone-bridge balance condition, or the potentiometer principle — applies directly to your figure too; only the specific labels may differ.

Formula Sheet

I. Electric Current and Current Density

Formula: Current and charge: $Q = It$; $n = \frac{Q}{e}$ (number of electrons)

Formula: Variable current: $Q = \int_{t_1}^{t_2} i dt = \text{area under the } I-t \text{ graph}$

II. Drift Velocity and Mobility

Formula: Microscopic form of current: $I = nAev_d \Rightarrow v_d = \frac{I}{nAe}$, $n = \text{free-electron density}$

III. Resistance and Resistivity

Formula: Ohm's Law: $V = IR \Rightarrow R = \frac{V}{I}$

Formula: Resistance in terms of resistivity: $R = \rho \frac{L}{A} \Rightarrow \rho = \frac{RA}{L}$; conductivity $\sigma = \frac{1}{\rho}$

Formula: Current density form: $J = \sigma E = \frac{E}{\rho} \Rightarrow \rho = \frac{E}{J}$

IV. Change in Resistance on Stretching the Wire

Formula: Volume conserved on stretching: $A_1L_1 = A_2L_2$; if length becomes n times, $R' = n^2R$

V. Effect of Temperature on Resistance and Resistivity

Formula: $R_T = R_0(1 + \alpha T)$, $\alpha = \frac{R_T - R_0}{R_0 T}$, $\alpha = \text{temperature coefficient of resistance}$

VI. Combination of Resistances

Formula: Series: $R_{eq} = R_1 + R_2 + \dots + R_n$; Parallel: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$

VII. Electric Power and Energy

Formula: $P = VI = I^2R = \frac{V^2}{R}$; Electrical energy: $E = Pt$

VIII. Electric Cell

Formula: Terminal PD (discharging): $V = \varepsilon - Ir \Rightarrow r = \frac{\varepsilon - V}{I}$; (charging): $V = \varepsilon + Ir$

IX. Combination of Cells

Formula: Series (n cells): $\varepsilon_{\text{total}} = \sum \varepsilon_i$, $r_{\text{total}} = \sum r_i$

Formula: Parallel (n identical cells): effective emf = ε , effective internal resistance = r/n

X. Kirchhoff's Laws

Formula: KCL (junction rule): $\sum I_{\text{in}} = \sum I_{\text{out}}$

Formula: KVL (loop rule): $\sum \varepsilon = \sum IR$ around any closed loop

XI. Wheatstone Bridge and Metre Bridge

Formula: Balanced Wheatstone bridge: $\frac{P}{Q} = \frac{R}{S}$ (no current through galvanometer)

Formula: Metre bridge: $\frac{R}{S} = \frac{l}{100 - l}$, l = balance length from left end

XII. Potentiometer

Formula: Potential gradient: $k = \frac{\varepsilon}{R_w + r} \times \frac{R_w}{L}$ (V per unit length)

Formula: Balance condition: $\text{emf} \propto \text{balance length}$, $\frac{\varepsilon_1}{\varepsilon_2} = \frac{l_1}{l_2}$

XIII. Miscellaneous

Formula: Combined applications of Ohm's Law, series-parallel reduction, and Kirchhoff's Laws as per the circuit topology

I. Electric Current and Current Density

Question 1

A current of $20 \mu\text{A}$ flows in a wire for 30 seconds. Find the amount of charge and the number of free electrons transferred through the wire.

Given: $I = 20 \times 10^{-6} \text{ A}$; $t = 30 \text{ s}$; $e = 1.6 \times 10^{-19} \text{ C}$

Formula: $Q = It$; $n = Q/e$

Solution:

$$Q = It = 20 \times 10^{-6} \times 30 = 6 \times 10^{-4} \text{ C}$$

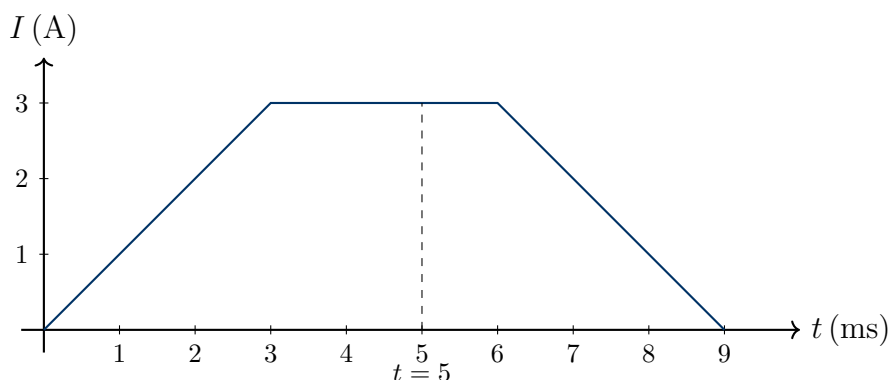
$$n = \frac{Q}{e} = \frac{6 \times 10^{-4}}{1.6 \times 10^{-19}} = 3.75 \times 10^{15} \text{ electrons}$$

Final Answer

$Q = 6 \times 10^{-4} \text{ C}$; $n = 3.75 \times 10^{15} \text{ electrons}$

Question 2

The current passing through a wire varies with time as shown in the graph (current rises linearly from 0 to 3 A during 0–3 ms, stays at 3 A during 3–6 ms, then falls linearly to 0 during 6–9 ms). Calculate the charge passing through the wire in (i) 0 s to 5 ms, (ii) 0 to 9 ms.



Current I versus time t for the wire (0–3 ms rise, 3–6 ms plateau, 6–9 ms fall)

Formula: $Q = \int I dt = \text{area under } I-t \text{ graph}$

Solution:

(i) 0 to 5 ms:

- Triangle (0–3 ms): area = $\frac{1}{2} \times 3 \times 10^{-3} \times 3 = 4.5 \times 10^{-3} \text{ C}$

- Rectangle (3–5 ms): area = $2 \times 10^{-3} \times 3 = 6 \times 10^{-3} \text{ C}$
- Total: $Q_1 = 4.5 + 6.0 = 10.5 \text{ mC}$

(ii) 0 to 9 ms:

- Triangle (0–3 ms): 4.5 mC
- Rectangle (3–6 ms): $3 \times 10^{-3} \times 3 = 9.0 \text{ mC}$
- Triangle (6–9 ms): $\frac{1}{2} \times 3 \times 10^{-3} \times 3 = 4.5 \text{ mC}$
- Total: $Q_2 = 4.5 + 9.0 + 4.5 = 18.0 \text{ mC}$

Final Answer

(i) $Q_1 = 10.5 \text{ mC}$; (ii) $Q_2 = 18 \text{ mC}$

Question 3

The current in a wire varies with time according to $i = 4 + 2t$, where i is in ampere and t in seconds. Calculate the quantity of charge which has passed through a cross-section of the wire during $t = 2 \text{ s}$ to $t = 6 \text{ s}$.

Given: $i = 4 + 2t$; $t_1 = 2 \text{ s}$; $t_2 = 6 \text{ s}$

Formula: $Q = \int_{t_1}^{t_2} i \, dt$

Solution:

$$Q = \int_2^6 (4 + 2t) \, dt = \left[4t + t^2 \right]_2^6 = (24 + 36) - (8 + 4) = 60 - 12 = 48 \text{ C}$$

Final Answer

$Q = 48 \text{ C}$

II. Drift Velocity and Mobility

Question 4

The cross-sectional area of a wire is $1.0 \times 10^{-7} \text{ m}^2$ and the density of free electrons is $2.0 \times 10^{28} \text{ m}^{-3}$. What will be the drift velocity of the free electrons for a current of 3.2 A in the wire?

Given: $A = 1.0 \times 10^{-7} \text{ m}^2$; $n = 2.0 \times 10^{28} \text{ m}^{-3}$; $I = 3.2 \text{ A}$; $e = 1.6 \times 10^{-19} \text{ C}$

Formula: $I = nAev_d \Rightarrow v_d = \frac{I}{nAe}$

Solution:

$$v_d = \frac{I}{nAe} = \frac{3.2}{2.0 \times 10^{28} \times 1.0 \times 10^{-7} \times 1.6 \times 10^{-19}} = \frac{3.2}{2.0 \times 10^{28} \times 1.6 \times 10^{-26}} = \frac{3.2}{320} = 1.0 \times 10^{-2} \text{ m s}^{-1}$$

Final Answer

$$v_d = 1.0 \times 10^{-2} \text{ m s}^{-1}$$

Question 5

A 3.0 m long copper wire is carrying a current of 3.0 A. How long does it take for an electron to drift from one end of the wire to the other? The cross-sectional area of the wire is $2.0 \times 10^{-6} \text{ m}^2$ and the number of conduction electrons in copper is $8.5 \times 10^{28} \text{ m}^{-3}$.

Given: $L = 3.0 \text{ m}$; $I = 3.0 \text{ A}$; $A = 2.0 \times 10^{-6} \text{ m}^2$; $n = 8.5 \times 10^{28} \text{ m}^{-3}$; $e = 1.6 \times 10^{-19} \text{ C}$

Formula: $v_d = \frac{I}{nAe}$; $t = L/v_d$

Solution:

$$v_d = \frac{3.0}{8.5 \times 10^{28} \times 2.0 \times 10^{-6} \times 1.6 \times 10^{-19}} = \frac{3.0}{8.5 \times 2.0 \times 1.6 \times 10^{28-6-19}} = \frac{3.0}{27.2 \times 10^3} = \frac{3.0}{2.72 \times 10^4} \approx$$

$$t = \frac{L}{v_d} = \frac{3.0}{1.103 \times 10^{-4}} \approx 2.72 \times 10^4 \text{ s} \approx 7.5 \text{ h}$$

Final Answer

$$t \approx 7.5 \text{ h}$$

★ Likely Exam Question

Question 6

90 C of charge flows for 75 minutes in a silver wire of diameter 1.0 mm. The free-electron density in the wire is $5.8 \times 10^{28} \text{ m}^{-3}$. Find (i) the current, (ii) the drift velocity of free electrons in the wire.

Given: $Q = 90 \text{ C}$; $t = 75 \times 60 = 4500 \text{ s}$; $d = 1.0 \text{ mm}$, so $r = 0.5 \times 10^{-3} \text{ m}$; $n = 5.8 \times 10^{28} \text{ m}^{-3}$; $e = 1.6 \times 10^{-19} \text{ C}$

Formula: $I = Q/t$; $A = \pi r^2$; $v_d = I/(nAe)$

Solution:

(i) **Current:**

$$I = \frac{Q}{t} = \frac{90}{4500} = 0.02 \text{ A}$$

(ii) **Cross-sectional area:**

$$\begin{aligned} A &= \pi r^2 = \pi \times (0.5 \times 10^{-3})^2 = \pi \times 2.5 \times 10^{-7} \approx 7.854 \times 10^{-7} \text{ m}^2 \\ v_d &= \frac{I}{nAe} = \frac{0.02}{5.8 \times 10^{28} \times 7.854 \times 10^{-7} \times 1.6 \times 10^{-19}} \\ &= \frac{0.02}{5.8 \times 7.854 \times 1.6 \times 10^{28-7-19}} = \frac{0.02}{72.885 \times 10^2} = \frac{0.02}{7288.5} \approx 2.74 \times 10^{-6} \text{ m s}^{-1} \end{aligned}$$

Final Answer

(i) $I = 0.02 \text{ A}$; (ii) $v_d \approx 2.7 \times 10^{-6} \text{ m s}^{-1}$

Question 7

Estimate the average drift speed of conduction electrons in a copper wire of cross-sectional area $1.0 \times 10^{-7} \text{ m}^2$ carrying a current of 1.5 A. Assume each copper atom contributes one conduction electron. Density of copper = $9.0 \times 10^3 \text{ kg m}^{-3}$, atomic mass = 63.5, Avogadro's number $N_A = 6.0 \times 10^{23} \text{ mol}^{-1}$.

Given: $A = 1.0 \times 10^{-7} \text{ m}^2$; $I = 1.5 \text{ A}$; $\rho_{\text{Cu}} = 9.0 \times 10^3 \text{ kg m}^{-3}$; $M = 63.5 \text{ g/mol}$; $N_A = 6.0 \times 10^{23} \text{ mol}^{-1}$; $e = 1.6 \times 10^{-19} \text{ C}$

Formula: $n = \frac{\rho_{\text{Cu}} \times N_A}{M}$; $v_d = \frac{I}{nAe}$

Solution:

$$\begin{aligned} n &= \frac{9.0 \times 10^3 \times 6.0 \times 10^{23}}{63.5 \times 10^{-3}} = \frac{5.4 \times 10^{27}}{6.35 \times 10^{-2}} = 8.504 \times 10^{28} \text{ m}^{-3} \\ v_d &= \frac{1.5}{8.504 \times 10^{28} \times 1.0 \times 10^{-7} \times 1.6 \times 10^{-19}} = \frac{1.5}{8.504 \times 1.6 \times 10^2} = \frac{1.5}{1360.6} \approx 1.1 \times 10^{-3} \text{ m s}^{-1} \end{aligned}$$

Final Answer

$$v_d \approx 1.1 \times 10^{-3} \text{ m s}^{-1}$$

III. Resistance and Resistivity

Question 8

Find out the resistivity of a conductor in which a current density of 2.5 A m^{-2} is found when an electric field of 15 V m^{-1} is applied on it.

Given: $J = 2.5 \text{ A m}^{-2}$; $E = 15 \text{ V m}^{-1}$

Formula: $J = \sigma E = E/\rho \Rightarrow \rho = E/J$

Solution:

$$\rho = \frac{E}{J} = \frac{15}{2.5} = 6 \text{ } \Omega\text{-m}$$

Final Answer

$$\rho = 6 \text{ } \Omega\text{-m}$$

Question 9

A very small current is sent through a wire of length 15 m and cross-sectional area $6.0 \times 10^{-7} \text{ m}^2$. Its resistance is measured to be $5.0 \text{ } \Omega$. Find the specific resistance of the material and the specific conductance.

Given: $L = 15 \text{ m}$; $A = 6.0 \times 10^{-7} \text{ m}^2$; $R = 5.0 \text{ } \Omega$

Formula: $R = \rho L/A \Rightarrow \rho = RA/L$; $\sigma = 1/\rho$

Solution:

$$\rho = \frac{RA}{L} = \frac{5.0 \times 6.0 \times 10^{-7}}{15} = \frac{3.0 \times 10^{-6}}{15} = 2.0 \times 10^{-7} \text{ } \Omega\text{-m}$$

$$\sigma = \frac{1}{\rho} = \frac{1}{2.0 \times 10^{-7}} = 5.0 \times 10^6 \text{ } (\Omega\text{-m})^{-1}$$

Final Answer

$$\rho = 2.0 \times 10^{-7} \text{ } \Omega\text{-m}; \quad \sigma = 5.0 \times 10^6 \text{ } (\Omega\text{-m})^{-1}$$

Question 10

A 200 V bulb takes a current of 0.2 A. The length of its filament is 22 cm and the specific resistance of the filament material is $7 \times 10^{-5} \text{ } \Omega\text{-cm}$. Find the radius of the filament ($\pi = 22/7$).

Given: $V = 200 \text{ V}$; $I = 0.2 \text{ A}$; $L = 22 \text{ cm} = 0.22 \text{ m}$; $\rho = 7 \times 10^{-5} \Omega\text{-cm} = 7 \times 10^{-7} \Omega\text{-m}$

Formula: $R = V/I$; $R = \rho L/A$; $A = \pi r^2$

Solution:

$$R = \frac{V}{I} = \frac{200}{0.2} = 1000 \Omega$$

$$A = \frac{\rho L}{R} = \frac{7 \times 10^{-7} \times 0.22}{1000} = \frac{1.54 \times 10^{-7}}{1000} = 1.54 \times 10^{-10} \text{ m}^2$$

$$r = \sqrt{\frac{A}{\pi}} = \sqrt{\frac{1.54 \times 10^{-10}}{22/7}} = \sqrt{\frac{1.54 \times 10^{-10} \times 7}{22}} = \sqrt{\frac{1.078 \times 10^{-9}}{22}} = \sqrt{4.9 \times 10^{-11}} = 7 \times 10^{-6} \text{ m} = 7 \times 10^{-4} \text{ cm}$$

Final Answer

Radius $r = 7 \times 10^{-4} \text{ cm}$

Question 11

A piece of copper of mass 50 g is drawn into a wire of length 50 m. Find the resistance of the wire. The density and specific resistance of copper are $8.9 \times 10^3 \text{ kg m}^{-3}$ and $1.7 \times 10^{-8} \Omega\text{-m}$ respectively.

Given: $m = 50 \text{ g} = 0.05 \text{ kg}$; $L = 50 \text{ m}$; $\rho_{\text{Cu}} = 8.9 \times 10^3 \text{ kg m}^{-3}$; $\rho_e = 1.7 \times 10^{-8} \Omega\text{-m}$

Formula: $V_{\text{ol}} = m/\rho_{\text{Cu}} = AL \Rightarrow A = m/(\rho_{\text{Cu}}L)$; $R = \rho_e L/A$

Solution:

$$A = \frac{m}{\rho_{\text{Cu}} \times L} = \frac{0.05}{8.9 \times 10^3 \times 50} = \frac{0.05}{4.45 \times 10^5} = 1.124 \times 10^{-7} \text{ m}^2$$

$$R = \frac{\rho_e L}{A} = \frac{1.7 \times 10^{-8} \times 50}{1.124 \times 10^{-7}} = \frac{8.5 \times 10^{-7}}{1.124 \times 10^{-7}} \approx 7.56 \Omega \approx 7.6 \Omega$$

Final Answer

$R \approx 7.6 \Omega$

Question 12

A rectangular block of iron has dimensions $1.2 \text{ cm} \times 1.2 \text{ cm} \times 15 \text{ cm}$. Compute the resistance of the block (a) between the two square ends, (b) between the two opposite rectangular faces. The specific resistance of iron at room temperature is $9.68 \times 10^{-8} \Omega\text{-m}$.

Given: Dimensions: $1.2 \text{ cm} \times 1.2 \text{ cm} \times 15 \text{ cm}$; $\rho = 9.68 \times 10^{-8} \Omega\text{-m}$

Formula: $R = \rho L/A$

Solution:

(a) **Between two square ends** ($L = 15 \text{ cm}$, $A = 1.2 \times 1.2 \text{ cm}^2$):

$$L_a = 15 \text{ cm} = 0.15 \text{ m}; \quad A_a = (1.2 \times 10^{-2})^2 = 1.44 \times 10^{-4} \text{ m}^2$$

$$R_a = \frac{9.68 \times 10^{-8} \times 0.15}{1.44 \times 10^{-4}} = \frac{1.452 \times 10^{-8}}{1.44 \times 10^{-4}} = 1.008 \times 10^{-4} \Omega \approx 100 \mu\Omega$$

(b) **Between two opposite rectangular faces** ($L = 1.2 \text{ cm}$, $A = 15 \times 1.2 \text{ cm}^2$):

$$L_b = 1.2 \times 10^{-2} \text{ m}; \quad A_b = 15 \times 10^{-2} \times 1.2 \times 10^{-2} = 1.8 \times 10^{-3} \text{ m}^2$$

$$R_b = \frac{9.68 \times 10^{-8} \times 1.2 \times 10^{-2}}{1.8 \times 10^{-3}} = \frac{1.1616 \times 10^{-9}}{1.8 \times 10^{-3}} = 6.453 \times 10^{-7} \Omega \approx 0.65 \mu\Omega$$

Final Answer

(a) $R_a \approx 100 \mu\Omega$; (b) $R_b \approx 0.65 \mu\Omega$

Question 13

What length of a wire of diameter 0.46 mm and specific resistance $50 \times 10^{-6} \Omega\text{-cm}$ will be required to prepare a coil of 10Ω ?

Given: $d = 0.46 \text{ mm} = 0.046 \text{ cm}$; $r = 0.023 \text{ cm}$; $\rho = 50 \times 10^{-6} \Omega\text{-cm}$; $R = 10 \Omega$

Formula: $R = \rho L/A \Rightarrow L = RA/\rho$; $A = \pi r^2$

Solution:

$$A = \pi r^2 = \frac{22}{7} \times (0.023)^2 = \frac{22}{7} \times 5.29 \times 10^{-4} = 3.14159 \times 5.29 \times 10^{-4} = 1.662 \times 10^{-3} \text{ cm}^2$$

$$L = \frac{RA}{\rho} = \frac{10 \times 1.662 \times 10^{-3}}{50 \times 10^{-6}} = \frac{1.662 \times 10^{-2}}{50 \times 10^{-6}} = \frac{1.662 \times 10^{-2}}{5 \times 10^{-5}} = 332 \text{ cm}$$

Final Answer

$L = 332 \text{ cm}$

Question 14

Two copper wires A and B of equal masses are taken. The length of A is double the length of B . If the resistance of wire A is 160Ω , calculate the resistance of wire B .

Given: $m_A = m_B$ (equal mass); $L_A = 2L_B$; $R_A = 160 \Omega$

Formula: $R = \rho L/A$; Equal mass $\Rightarrow \rho_{\text{Cu}} A_A L_A = \rho_{\text{Cu}} A_B L_B \Rightarrow A_A/A_B = L_B/L_A = 1/2$; $R \propto L/A$

Solution:

Since mass is equal and material is same:

$$A_A L_A = A_B L_B \implies A_A = A_B \times \frac{L_B}{L_A} = A_B \times \frac{1}{2}$$

$$\frac{R_A}{R_B} = \frac{L_A/A_A}{L_B/A_B} = \frac{L_A}{L_B} \times \frac{A_B}{A_A} = 2 \times 2 = 4$$

$$R_B = \frac{R_A}{4} = \frac{160}{4} = 40 \, \Omega$$

Final Answer

$$R_B = 40 \, \Omega$$

Question 15

Two wires of different materials; of lengths 50 cm and 75 cm and respective diameters 0.2 mm and 0.1 mm, have resistances of 0.8 Ω and 1.5 Ω respectively. What is the ratio of the specific resistances of the materials?

Given: $L_1 = 50$ cm, $d_1 = 0.2$ mm, $R_1 = 0.8 \, \Omega$; $L_2 = 75$ cm, $d_2 = 0.1$ mm, $R_2 = 1.5 \, \Omega$

Formula: $\rho = RA/L = R\pi(d/2)^2/L$; $\rho \propto Rd^2/L$

Solution:

$$\begin{aligned} \frac{\rho_1}{\rho_2} &= \frac{R_1 d_1^2 / L_1}{R_2 d_2^2 / L_2} = \frac{R_1}{R_2} \times \frac{d_1^2}{d_2^2} \times \frac{L_2}{L_1} = \frac{0.8}{1.5} \times \frac{(0.2)^2}{(0.1)^2} \times \frac{75}{50} \\ &= \frac{0.8}{1.5} \times 4 \times 1.5 = \frac{0.8 \times 4 \times 1.5}{1.5} = 0.8 \times 4 = \frac{16}{5} \end{aligned}$$

So $\rho_1 : \rho_2 = 16 : 5$.

Final Answer

$$\rho_1 : \rho_2 = 16 : 5$$

IV. Change in Resistance on Stretching the Wire

Question 16

A thick wire of $10\ \Omega$ resistance is drawn into a thin wire so that its length becomes three times. Calculate its new resistance.

Given: $R = 10\ \Omega$; length becomes $3L$

Formula: Volume constant: $AL = A'(3L) \Rightarrow A' = A/3$; $R' = \rho(3L)/A' = \rho(3L)/(A/3) = 9\rho L/A = 9R$; alternatively $R' = n^2 R$

Solution:

When a wire is stretched to $n = 3$ times its original length, volume is conserved:

$$R' = n^2 R = 3^2 \times 10 = 9 \times 10 = 90\ \Omega$$

Final Answer

$$R' = 90\ \Omega$$

Question 17

A 1-kg piece of copper is drawn into a wire 1 mm thick and another piece (of the same mass) into a wire 2 mm thick. Compare the resistances of these wires.

Given: Same mass, same material; $d_1 = 1\ \text{mm}$, $d_2 = 2\ \text{mm}$

Formula: $V_{\text{ol}} = AL = \text{const}$; $L \propto 1/A \propto 1/d^2$; $R = \rho L/A \propto L/A \propto L \times (1/d^2)$

Solution:

Since volume is the same:

$$A_1 L_1 = A_2 L_2 \Rightarrow \frac{L_1}{L_2} = \frac{A_2}{A_1} = \frac{d_2^2}{d_1^2} = \frac{4}{1}$$

$$\frac{R_1}{R_2} = \frac{L_1/A_1}{L_2/A_2} = \frac{L_1}{L_2} \times \frac{A_2}{A_1} = 4 \times 4 = 16$$

Final Answer

$$R_1 : R_2 = 16 : 1$$

V. Effect of Temperature on Resistance and Resistivity

Question 18

The resistance of a silver wire at 0°C is $1.25\ \Omega$. Up to what temperature must it be heated so that its resistance is doubled? The temperature coefficient of resistance of silver is $0.00375\ ^\circ\text{C}^{-1}$. Will the temperature be the same for all conductors of silver?

Given: $R_0 = 1.25\ \Omega$; $R_T = 2R_0 = 2.50\ \Omega$; $\alpha = 0.00375\ ^\circ\text{C}^{-1}$

Formula: $R_T = R_0(1 + \alpha T)$

Solution:

$$2R_0 = R_0(1 + \alpha T) \implies 2 = 1 + \alpha T \implies \alpha T = 1$$

$$T = \frac{1}{\alpha} = \frac{1}{0.00375} = 266.7^\circ\text{C} \approx 267^\circ\text{C}$$

Yes, the temperature will be the same for all silver conductors regardless of dimensions, since α is a material property.

Final Answer

$T \approx 267^\circ\text{C}$; Yes, same for all silver conductors

Question 19

The resistance of the coil used in a platinum-resistance thermometer at 0°C is $3.00\ \Omega$ and at 100°C is $3.75\ \Omega$. Its resistance at an unknown temperature is measured at $3.15\ \Omega$. Calculate the unknown temperature.

Given: $R_0 = 3.00\ \Omega$; $R_{100} = 3.75\ \Omega$; $R_T = 3.15\ \Omega$

Formula: $\alpha = (R_{100} - R_0)/(R_0 \times 100)$; $T = (R_T - R_0)/(\alpha R_0)$

Solution:

$$\alpha = \frac{R_{100} - R_0}{R_0 \times 100} = \frac{3.75 - 3.00}{3.00 \times 100} = \frac{0.75}{300} = 2.5 \times 10^{-3}\ ^\circ\text{C}^{-1}$$

$$T = \frac{R_T - R_0}{\alpha R_0} = \frac{3.15 - 3.00}{2.5 \times 10^{-3} \times 3.00} = \frac{0.15}{7.5 \times 10^{-3}} = 20^\circ\text{C}$$

Final Answer

$T = 20^\circ\text{C}$

Question 20

A metallic wire has a resistance of $3.0 \, \Omega$ at 0°C and $4.8 \, \Omega$ at 150°C . Find the temperature coefficient of resistance of its material.

Given: $R_0 = 3.0 \, \Omega$; $R_{150} = 4.8 \, \Omega$; $T = 150^\circ\text{C}$

Formula: $R_T = R_0(1 + \alpha T)$; $\alpha = (R_T - R_0)/(R_0 T)$

Solution:

$$\alpha = \frac{R_T - R_0}{R_0 T} = \frac{4.8 - 3.0}{3.0 \times 150} = \frac{1.8}{450} = 4 \times 10^{-3} \, ^\circ\text{C}^{-1}$$

Final Answer

$$\alpha = 4 \times 10^{-3} \, ^\circ\text{C}^{-1}$$

Question 21

The resistance of a carbon filament at 0°C is $104 \, \Omega$. What should be the resistance of an iron filament which should be connected in series with the carbon filament so that the resistance of the combination does not change with temperature? The temperature coefficient of resistance of iron $\alpha_{\text{Fe}} = +0.0052 \, ^\circ\text{C}^{-1}$ and of carbon $\alpha_C = -0.0003 \, ^\circ\text{C}^{-1}$.

Given: $R_C = 104 \, \Omega$; $\alpha_C = -0.0003 \, ^\circ\text{C}^{-1}$; $\alpha_{\text{Fe}} = +0.0052 \, ^\circ\text{C}^{-1}$

Formula: For zero net change: $R_C \alpha_C + R_{\text{Fe}} \alpha_{\text{Fe}} = 0 \Rightarrow R_{\text{Fe}} = -R_C \alpha_C / \alpha_{\text{Fe}}$

Solution:

For the combination resistance to not change with temperature, the total temperature coefficient must be zero:

$$R_C \alpha_C + R_{\text{Fe}} \alpha_{\text{Fe}} = 0$$

$$R_{\text{Fe}} = \frac{-R_C \alpha_C}{\alpha_{\text{Fe}}} = \frac{-104 \times (-0.0003)}{0.0052} = \frac{0.0312}{0.0052} = 6.00 \, \Omega$$

Final Answer

$$R_{\text{Fe}} = 6.00 \, \Omega \text{ at } 0^\circ\text{C}$$

VI. Combination of Resistances

Question 22

How can three resistances of $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ be connected to give an equivalent resistance of $4\ \Omega$?

Solution:

Connect $3\ \Omega$ and $6\ \Omega$ in parallel, then connect that combination in series with $2\ \Omega$:

Step 1 — parallel combination of $3\ \Omega$ and $6\ \Omega$:

$$R_{36} = \frac{3 \times 6}{3 + 6} = \frac{18}{9} = 2\ \Omega$$

Step 2 — series with $2\ \Omega$:

$$R_{\text{eq}} = 2 + 2 = 4\ \Omega \quad \checkmark$$

Final Answer

Connect $3\ \Omega$ and $6\ \Omega$ in parallel; connect result in series with $2\ \Omega$ to get $4\ \Omega$.

Question 23

Three resistors are joined to form a triangle ABC . The resistances of sides AB , BC and CA are $40\ \Omega$, $60\ \Omega$ and $100\ \Omega$ respectively. Find the equivalent resistance between points A and B .

Given: $R_{AB} = 40\ \Omega$; $R_{BC} = 60\ \Omega$; $R_{CA} = 100\ \Omega$

Solution:

Between A and B : the direct path is $R_{AB} = 40\ \Omega$; the indirect path goes $A \rightarrow C \rightarrow B$, i.e., $R_{CA} + R_{BC} = 100 + 60 = 160\ \Omega$.

These two paths are in parallel:

$$R_{\text{eq}} = \frac{40 \times 160}{40 + 160} = \frac{6400}{200} = 32\ \Omega$$

Final Answer

$$R_{\text{eq}} = 32\ \Omega$$

Question 24

Calculate the equivalent resistance between points A and B in each of the shown networks of resistors.

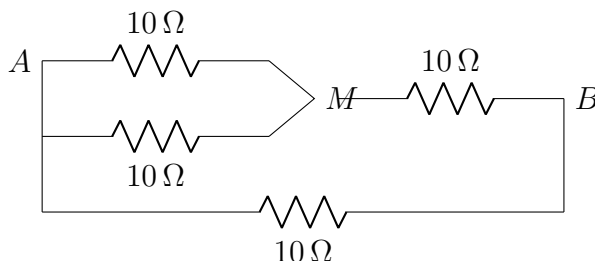
Network (a): Four $10\ \Omega$ resistors arranged in a symmetric ladder — one $10\ \Omega$ in series on each arm and two $10\ \Omega$ across the middle.

Network (b): Six $10\ \Omega$ resistors arranged with three in series on the top row and

three in the bottom row connected in a chain.

Solution:

Network (a):



Two of the $10\ \Omega$ resistors run side by side from A to the junction M — they are in **parallel**:

$$R_{AM} = \frac{10 \times 10}{10 + 10} = 5\ \Omega$$

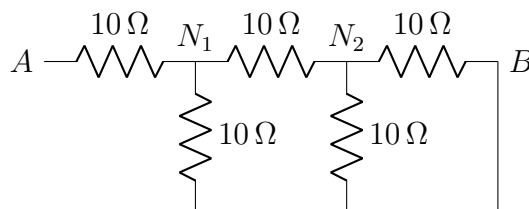
This is in **series** with the $10\ \Omega$ resistor from M to B , giving the resistance of the upper path:

$$R_{\text{upper}} = 5 + 10 = 15\ \Omega$$

This upper path is in **parallel** with the fourth $10\ \Omega$ resistor connected directly across A and B :

$$R_{AB}^{(a)} = \frac{15 \times 10}{15 + 10} = \frac{150}{25} = 6\ \Omega$$

Network (b):



This is a **ladder network** — reduce it step by step starting from the end *farthest* from A .

Step 1: The right-most "rung" (N_2 down to the bottom rail, which connects straight across to B) is in parallel with the right-most top resistor (N_2 to B):

$$10 \parallel 10 = 5\ \Omega$$

Step 2: Add this in series with the middle top resistor (N_1 to N_2):

$$5 + 10 = 15\ \Omega$$

Step 3: This $15\ \Omega$ is now in parallel with the rung at N_1 (which also returns to B along the bottom rail):

$$15 \parallel 10 = \frac{150}{25} = 6\ \Omega$$

Step 4: Finally, add the first top resistor (A to N_1) in series:

$$R_{AB}^{(b)} = 6 + 10 = 16 \, \Omega$$

Final Answer

$$(a) R_{AB} = 6 \, \Omega; \quad (b) R_{AB} = 16 \, \Omega$$

Question 25

A $5 \, \Omega$ resistor is connected in series with a parallel combination of n resistors each of $6 \, \Omega$. The equivalent resistance is $7 \, \Omega$. Find n .

Given: Series resistor = $5 \, \Omega$; n resistors of $6 \, \Omega$ in parallel; $R_{eq} = 7 \, \Omega$

Formula: $R_{parallel} = 6/n$; $R_{eq} = 5 + 6/n = 7$

Solution:

$$5 + \frac{6}{n} = 7 \implies \frac{6}{n} = 2 \implies n = 3$$

Final Answer

$$n = 3$$

Question 26

Four resistors, each $10 \, \Omega$, are joined in the form of a square. Find the equivalent resistance between two diagonally-opposite corners.

Solution:

Label corners A, B, C, D clockwise. Measuring between A (top-left) and C (bottom-right):

Top path: $A \rightarrow B \rightarrow C = 10 + 10 = 20 \, \Omega$

Bottom path: $A \rightarrow D \rightarrow C = 10 + 10 = 20 \, \Omega$

These two paths are in parallel:

$$R_{AC} = \frac{20 \times 20}{20 + 20} = \frac{400}{40} = 10 \, \Omega$$

Final Answer

$$R_{eq} = 10 \, \Omega$$

Question 27

Join three $3 \, \Omega$ resistances to obtain an effective resistance of $2 \, \Omega$.

Solution:

Connect two of the $3\ \Omega$ resistors in series with each other, then connect that combination in parallel with the third $3\ \Omega$ resistor.

Series pair: $3 + 3 = 6\ \Omega$

Parallel with third $3\ \Omega$:

$$R_{\text{eq}} = \frac{6 \times 3}{6 + 3} = \frac{18}{9} = 2\ \Omega \quad \checkmark$$

Final Answer

Two in series, and the third in parallel with them $\Rightarrow R_{\text{eq}} = 2\ \Omega$

Question 28

A wire of resistance $0.1\ \Omega/\text{cm}$ is bent in the form of a square $ABCD$ of side $10\ \text{cm}$. An exactly similar wire is connected across points B and D . Find (a) the equivalent resistance between the other two corners A and C , (b) if a $2\ \text{V}$ battery is connected between A and C , the current drawn and ammeter reading across B and D .

Given: Resistance per cm = $0.1\ \Omega/\text{cm}$; side = $10\ \text{cm}$

Solution:

Resistance of each side = $0.1 \times 10 = 1\ \Omega$. Resistance of wire $BD = 0.1 \times 10 = 1\ \Omega$ (diagonal).

(a) From A to C :

- Path $A \rightarrow B \rightarrow C = 1 + 1 = 2\ \Omega$
- Path $A \rightarrow D \rightarrow C = 1 + 1 = 2\ \Omega$
- The BD wire connects B and D ; by symmetry, $V_B = V_D$ when current enters at A and exits at C .

Since $V_B = V_D$, no current flows through the BD wire when voltage applied across AC . So the BD wire is irrelevant for the A – C resistance.

$$R_{AC} = \frac{2 \times 2}{2 + 2} = 1\ \Omega$$

(b) Current from battery ($V = 2\ \text{V}$, $R_{AC} = 1\ \Omega$):

$$I = \frac{V}{R_{AC}} = \frac{2}{1} = 2\ \text{A}$$

Ammeter across BD : Since $V_B = V_D$ by symmetry, the ammeter reads zero.

Final Answer

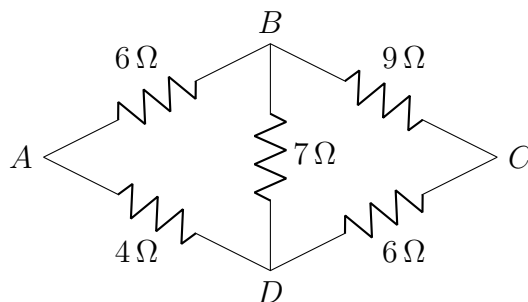
(a) $R_{AC} = 1\ \Omega$; (b) Current from battery = $2\ \text{A}$; Ammeter reads = 0

Question 29 (ISC 2000)

Calculate the equivalent resistance between points A and B in each of the following networks of resistors: **(a)** Diamond/bridge network with $6, 7, 4, 6, 9 \Omega$ resistors. **(b)** Two-rung ladder with $1, 10, 4, 2 \Omega$ resistors. **(c)** Three resistors of 6Ω each in the circuit.

Solution:

(a) Bridge (Wheatstone) network:



Check whether the bridge is *balanced*: $R_{AB} \cdot R_{DC} \stackrel{?}{=} R_{AD} \cdot R_{BC}$, i.e. $6 \times 6 = 36$ and $4 \times 9 = 36$. **Yes — balanced.**

A balanced Wheatstone bridge carries *no current* through its central arm (the 7Ω resistor), so we remove it:

$$R_{\text{top}} = R_{AB} + R_{BC} = 6 + 9 = 15 \Omega$$

$$R_{\text{bot}} = R_{AD} + R_{DC} = 4 + 6 = 10 \Omega$$

$$R_{AC}^{(a)} = \frac{15 \times 10}{15 + 10} = \frac{150}{25} = 6 \Omega$$

(b) Ladder network with $1, 10, 4, 2 \Omega$:

The ladder has three parallel branches between nodes A and B ; reduce from the far end toward A . First combine the 10Ω rung and 2Ω series resistor in parallel:

$$R_{10 \parallel 2} = \frac{10 \times 2}{10 + 2} = \frac{20}{12} = \frac{5}{3} \Omega$$

Add the 4Ω in series:

$$R_{\text{right}} = 4 + \frac{5}{3} = \frac{17}{3} \Omega$$

Finally place this in series with the 1Ω top resistor:

$$R_{AB}^{(b)} = 1 + \frac{17}{3} = \frac{3 + 17}{3} = \frac{20}{3} \Omega \approx 6.67 \Omega$$

(c) Two of the 6Ω resistors are in **series** (giving 12Ω), and this combination is in **parallel** with the third 6Ω :

$$R_{AB}^{(c)} = \frac{12 \times 6}{12 + 6} = \frac{72}{18} = 4 \Omega$$

Final Answer

$$(a) R_{AC} = 6 \Omega; \quad (b) R_{AB} = \frac{20}{3} \Omega \approx 6.67 \Omega; \quad (c) R_{AB} = 4 \Omega$$

VII. Electric Power and Energy

Question 30

How much charge flows through a 250 V–1000 W heater in one minute?

Given: $V = 250 \text{ V}$; $P = 1000 \text{ W}$; $t = 60 \text{ s}$

Formula: $I = P/V$; $Q = It$

Solution:

$$I = \frac{P}{V} = \frac{1000}{250} = 4 \text{ A}$$
$$Q = It = 4 \times 60 = 240 \text{ C}$$

Final Answer

$Q = 240 \text{ C}$

Question 31

Compute the voltage drop across a 2 kW electric heater element whose resistance when hot is 20Ω .

Given: $P = 2000 \text{ W}$; $R = 20 \Omega$

Formula: $P = V^2/R \Rightarrow V = \sqrt{PR}$

Solution:

$$V = \sqrt{PR} = \sqrt{2000 \times 20} = \sqrt{40000} = 200 \text{ V}$$

Final Answer

Voltage drop $V = 200 \text{ V}$

Question 32

An electric motor of 1 HP (horsepower) is running at 220 V. What is the current in the motor?

Given: $P = 1 \text{ HP} = 746 \text{ W}$; $V = 220 \text{ V}$

Formula: $I = P/V$

Solution:

$$I = \frac{P}{V} = \frac{746}{220} \approx 3.39 \text{ A}$$

Final Answer

$$I \approx 3.39 \text{ A}$$

Question 33

An electric press has a resistance of 44Ω and a current of 5.0 A flows through it. Calculate the power consumed by the press.

Given: $R = 44 \Omega$; $I = 5.0 \text{ A}$

Formula: $P = I^2 R$

Solution:

$$P = I^2 R = (5.0)^2 \times 44 = 25 \times 44 = 1100 \text{ W}$$

Final Answer

$$P = 1100 \text{ W}$$

Question 34

A current of 4.0 A flows in a heating-wire of 100 W . Determine the resistance of the wire.

Given: $I = 4.0 \text{ A}$; $P = 100 \text{ W}$

Formula: $P = I^2 R \Rightarrow R = P/I^2$

Solution:

$$R = \frac{P}{I^2} = \frac{100}{(4.0)^2} = \frac{100}{16} = 6.25 \Omega$$

Final Answer

$$R = 6.25 \Omega$$

Question 35

An electric heater has a rating of 1100 W – 220 V . How much electrical energy (in kW-h) will it consume in 4 hours? What is the resistance of its element?

Given: $P = 1100 \text{ W}$; $V = 220 \text{ V}$; $t = 4 \text{ h}$

Formula: $E = Pt$; $R = V^2/P$

Solution:

$$E = 1100 \text{ W} \times 4 \text{ h} = 4400 \text{ W-h} = 4.4 \text{ kW-h}$$

$$R = \frac{V^2}{P} = \frac{(220)^2}{1100} = \frac{48400}{1100} = 44 \Omega$$

Final Answer

$$E = 4.4 \text{ kW-h}; \quad R = 44 \Omega$$

Question 36

A heating element is marked 210 V, 630 W. What is the current drawn by the element when connected to 210 V DC mains? What is the resistance of the element?

Given: $V = 210 \text{ V}$; $P = 630 \text{ W}$

Formula: $I = P/V$; $R = V^2/P$

Solution:

$$I = \frac{P}{V} = \frac{630}{210} = 3.0 \text{ A}$$

$$R = \frac{V^2}{P} = \frac{(210)^2}{630} = \frac{44100}{630} = 70 \Omega$$

Final Answer

$$I = 3.0 \text{ A}; \quad R = 70 \Omega$$

Question 37

Find the resistance of the filament of a 100 W–220 V bulb when glowing. The length and radius of the filament are 0.1 m and $2.5 \times 10^{-5} \text{ m}$. Find the specific resistance of the material of the filament.

Given: $P = 100 \text{ W}$; $V = 220 \text{ V}$; $L = 0.1 \text{ m}$; $r = 2.5 \times 10^{-5} \text{ m}$

Formula: $R = V^2/P$; $\rho = RA/L = R\pi r^2/L$

Solution:

$$R = \frac{V^2}{P} = \frac{(220)^2}{100} = \frac{48400}{100} = 484 \Omega$$

$$A = \pi r^2 = 3.14159 \times (2.5 \times 10^{-5})^2 = 3.14159 \times 6.25 \times 10^{-10} = 1.963 \times 10^{-9} \text{ m}^2$$

$$\rho = \frac{RA}{L} = \frac{484 \times 1.963 \times 10^{-9}}{0.1} = \frac{9.501 \times 10^{-7}}{0.1} = 9.5 \times 10^{-6} \Omega\text{-m}$$

Final Answer

$$R = 484 \, \Omega; \quad \rho = 9.5 \times 10^{-6} \, \Omega\text{-m}$$

Question 38

Find the resistance of a 240 V–200 W electric bulb when glowing. If this resistance is 10 times the resistance at 0°C and the temperature of the glowing filament is 2000°C, find the temperature coefficient of resistance of the filament.

Given: $V = 240 \, \text{V}; \quad P = 200 \, \text{W}; \quad R_{2000} = 10R_0; \quad T = 2000 \, ^\circ\text{C}$

Formula: $R = V^2/P; \quad R_T = R_0(1 + \alpha T)$

Solution:

$$R_{2000} = \frac{V^2}{P} = \frac{(240)^2}{200} = \frac{57600}{200} = 288 \, \Omega$$

$$R_0 = \frac{R_{2000}}{10} = \frac{288}{10} = 28.8 \, \Omega$$

$$R_{2000} = R_0(1 + \alpha \times 2000) \implies 10 = 1 + 2000\alpha \implies \alpha = \frac{9}{2000} = 4.5 \times 10^{-3} \, ^\circ\text{C}^{-1}$$

Final Answer

$$R_{2000} = 288 \, \Omega; \quad \alpha = 4.5 \times 10^{-3} \, ^\circ\text{C}^{-1}$$

Question 39

A lamp marked '15 V–20 W' is connected to a DC source of 25 V and a resistance R in series for full illumination. Find (i) the current in the circuit, (ii) the resistance of the lamp, (iii) the resistance R .

Given: Lamp: $V_L = 15 \, \text{V}, P = 20 \, \text{W}; \quad \text{Supply: } V_S = 25 \, \text{V}; \quad R \text{ in series}$

Formula: $I = P/V_L; \quad R_{\text{lamp}} = V_L/I; \quad R = (V_S - V_L)/I$

Solution:

(i)

$$I = \frac{P}{V_L} = \frac{20}{15} = \frac{4}{3} \, \text{A}$$

(ii)

$$R_{\text{lamp}} = \frac{V_L}{I} = \frac{15}{4/3} = 15 \times \frac{3}{4} = \frac{45}{4} \, \Omega$$

(iii)

$$R = \frac{V_S - V_L}{I} = \frac{25 - 15}{4/3} = \frac{10 \times 3}{4} = \frac{30}{4} = 7.5 \, \Omega$$

Final Answer

(i) $I = 4/3 \text{ A}$; (ii) $R_{\text{lamp}} = 45/4 \text{ } \Omega$; (iii) $R = 7.5 \text{ } \Omega$

Question 40

We have a 30 W–6 V bulb which we want to glow by a supply of 120 V. What will have to be done for it?

Given: Bulb: $P = 30 \text{ W}$, $V_b = 6 \text{ V}$; Supply $V_S = 120 \text{ V}$

Solution:

Current needed by bulb: $I = P/V_b = 30/6 = 5 \text{ A}$.

Voltage to be dropped by series resistance: $V_R = V_S - V_b = 120 - 6 = 114 \text{ V}$.

$$R = \frac{V_R}{I} = \frac{114}{5} = 22.8 \text{ } \Omega$$

A resistance wire of $22.8 \text{ } \Omega$ should be connected in series with the bulb.

Final Answer

Connect a resistance of $22.8 \text{ } \Omega$ in series with the bulb.

Question 41

Two cities are 150 km away from each other. Electric power is transmitted through copper wires from one city to the other. There is a potential drop of 8 V/km in the wires and the average resistance of the wires is $0.5 \text{ } \Omega/\text{km}$. (i) Calculate the current flowing in the wires. (ii) Find out the loss of electric power in the wires.

Given: Distance = 150 km; potential drop = 8 V/km; $R = 0.5 \text{ } \Omega/\text{km}$

Formula: $V_{\text{drop}} = I \times R_{\text{total}}$; $P_{\text{loss}} = I^2 R_{\text{total}}$

Solution:

Total potential drop over 150 km: $V = 8 \times 150 = 1200 \text{ V}$

Total resistance: $R_{\text{total}} = 0.5 \times 150 = 75 \text{ } \Omega$

(i)

$$I = \frac{V}{R_{\text{total}}} = \frac{1200}{75} = 16 \text{ A}$$

(ii)

$$P_{\text{loss}} = I^2 R_{\text{total}} = (16)^2 \times 75 = 256 \times 75 = 19200 \text{ W} = 19.2 \text{ kW}$$

Final Answer

(i) $I = 16 \text{ A}$; (ii) $P_{\text{loss}} = 19.2 \text{ kW}$

Question 42

In a house with 220 V line the following three appliances are operating: (i) a 60 W bulb, (ii) a 1000 W heater and (iii) a 40 W radio. Calculate (a) the current drawn by the heater and (b) the current passing through the fuse for this line.

Given: $V = 220 \text{ V}$; $P_{\text{bulb}} = 60 \text{ W}$; $P_{\text{heater}} = 1000 \text{ W}$; $P_{\text{radio}} = 40 \text{ W}$; all in parallel

Formula: $I = P/V$; $I_{\text{fuse}} = I_{\text{total}} = \sum P_i/V$

Solution:

(a) Current drawn by heater:

$$I_{\text{heater}} = \frac{P_{\text{heater}}}{V} = \frac{1000}{220} = \frac{50}{11} \approx 4.55 \text{ A}$$

(b) Total current (all parallel):

$$I_{\text{total}} = \frac{P_{\text{bulb}} + P_{\text{heater}} + P_{\text{radio}}}{V} = \frac{60 + 1000 + 40}{220} = \frac{1100}{220} = 5.0 \text{ A}$$

Final Answer

(a) $I_{\text{heater}} = 50/11 \text{ A} \approx 4.55 \text{ A}$; (b) $I_{\text{fuse}} = 5.0 \text{ A}$

Question 43

Calculate the number of electrons drifting per second through the filament of a 100 W-220 V bulb, when glowing ($e = 1.6 \times 10^{-19} \text{ C}$).

Given: $P = 100 \text{ W}$; $V = 220 \text{ V}$; $e = 1.6 \times 10^{-19} \text{ C}$

Formula: $I = P/V$; $n = I/e$

Solution:

$$I = \frac{P}{V} = \frac{100}{220} = \frac{5}{11} \text{ A}$$

$$n = \frac{I}{e} = \frac{5/11}{1.6 \times 10^{-19}} = \frac{0.4545}{1.6 \times 10^{-19}} = 2.84 \times 10^{18} \text{ electrons per second}$$

Final Answer

$n \approx 2.84 \times 10^{18}$ electrons per second

Question 44

Two resistors $R_1 = 60 \, \Omega$ and $R_2 = 90 \, \Omega$ are connected in parallel. If the electric power consumed by R_1 is 15 W, calculate the power consumed by R_2 .

Given: $R_1 = 60 \, \Omega$, $P_1 = 15 \, \text{W}$; $R_2 = 90 \, \Omega$; parallel connection (same voltage)

Formula: $P = V^2/R$; In parallel: V same for both

Solution:

$$V^2 = P_1 R_1 = 15 \times 60 = 900 \, \text{V}^2$$

$$P_2 = \frac{V^2}{R_2} = \frac{900}{90} = 10 \, \text{W}$$

Final Answer

$$P_2 = 10 \, \text{W}$$

Question 45

Find the ratio of the resistances of a 40 W bulb and a 100 W bulb designed to work at 220 V supply line.

Given: $P_1 = 40 \, \text{W}$, $P_2 = 100 \, \text{W}$; $V = 220 \, \text{V}$ (same)

Formula: $R = V^2/P$; $R \propto 1/P$

Solution:

$$\frac{R_{40}}{R_{100}} = \frac{P_{100}}{P_{40}} = \frac{100}{40} = \frac{5}{2}$$

Final Answer

$$R_{40} : R_{100} = 5 : 2$$

Question 46

A 60 W bulb is switched on in a room. A 240 W heater is also turned on in the same room. The voltage of the mains is 120 V and the resistance of the connecting leads is $6 \, \Omega$. What is the change in the voltage at the bulb when the heater is turned on?

Given: $P_{\text{bulb}} = 60 \, \text{W}$, so $R_{\text{bulb}} = V^2/P = (120)^2/60 = 240 \, \Omega$; $P_{\text{heater}} = 240 \, \text{W}$, so $R_{\text{heater}} = (120)^2/240 = 60 \, \Omega$; $R_{\text{leads}} = 6 \, \Omega$; $V_{\text{mains}} = 120 \, \text{V}$

Solution:

Only bulb on:

$$I_1 = \frac{120}{240 + 6} = \frac{120}{246} \approx 0.4878 \text{ A}$$

$$V_{\text{bulb},1} = I_1 \times 240 = 0.4878 \times 240 \approx 117.1 \text{ V}$$

Or using the voltage divider: $V_{\text{bulb},1} = 120 \times \frac{240}{240+6} = 120 \times \frac{240}{246} = \frac{120 \times 240}{246} \approx 117.1 \text{ V}$.

Both on (bulb and heater in parallel):

$$R_{\text{parallel}} = \frac{240 \times 60}{240 + 60} = \frac{14400}{300} = 48 \Omega$$

$$I_2 = \frac{120}{48 + 6} = \frac{120}{54} = \frac{20}{9} \text{ A}$$

$$V_{\text{bulb},2} = I_2 \times 48 = \frac{20}{9} \times 48 = \frac{960}{9} \approx 106.7 \text{ V}$$

$$\Delta V = V_{\text{bulb},2} - V_{\text{bulb},1} \approx 106.7 - 117.1 = -10.4 \text{ V}$$

Final Answer

Voltage at bulb reduces by approximately 10.4 V when heater is turned on.

Question 47

A 500 W electric heater is designed to work with a 115 V line. If the voltage of the line drops to 110 V, what will be the percentage loss of energy liberated?

Given: $P_0 = 500 \text{ W}$ at $V_0 = 115 \text{ V}$; $V = 110 \text{ V}$; $R = \text{const}$

Formula: $P = V^2/R$; $P \propto V^2$

Solution:

$$\frac{P}{P_0} = \frac{V^2}{V_0^2} = \frac{(110)^2}{(115)^2} = \frac{12100}{13225} = 0.9149$$

$$\% \text{ loss} = \left(1 - \frac{P}{P_0}\right) \times 100 = (1 - 0.9149) \times 100 = 8.51\% \approx 8.5\%$$

Final Answer

Percentage loss $\approx 8.5\%$

VIII. Electric Cell

Question 48

The potential difference of a cell in an open-circuit is 6 V which falls to 4 V when a current of 2 A is drawn from the cell. Calculate the emf and internal resistance of the cell.

Given: $V_{\text{open}} = \varepsilon = 6 \text{ V}$; $V_{\text{load}} = 4 \text{ V}$; $I = 2 \text{ A}$

Formula: $V = \varepsilon - Ir \Rightarrow r = (\varepsilon - V)/I$

Solution:

$$r = \frac{\varepsilon - V}{I} = \frac{6 - 4}{2} = \frac{2}{2} = 1 \Omega$$

Final Answer

emf $\varepsilon = 6 \text{ V}$; internal resistance $r = 1 \Omega$

Question 49

The emf of a cell is 2.0 V. When it is connected to a 25Ω resistor, the terminal voltage of the cell is found to be 1.8 V. What is the internal resistance of the cell and the current in the 25Ω resistor?

Given: $\varepsilon = 2.0 \text{ V}$; $R = 25 \Omega$; $V_T = 1.8 \text{ V}$

Formula: $I = V_T/R$; $r = (\varepsilon - V_T)/I$

Solution:

$$I = \frac{V_T}{R} = \frac{1.8}{25} = 0.072 \text{ A}$$

$$r = \frac{\varepsilon - V_T}{I} = \frac{2.0 - 1.8}{0.072} = \frac{0.2}{0.072} \approx 2.78 \Omega$$

Final Answer

$r \approx 2.78 \Omega$; $I = 0.072 \text{ A}$

Question 50

A 200Ω resistor, a voltmeter and a 2.0 V cell of negligible internal resistance are connected in series. The voltmeter reads 1.0 V. Find the resistance of the voltmeter and the current in the circuit.

Given: $R = 200 \Omega$; $\varepsilon = 2.0 \text{ V}$; $r \approx 0$; $V_V = 1.0 \text{ V}$

Solution:

Voltage across R : $V_R = 2.0 - 1.0 = 1.0 \text{ V}$.

Current: $I = V_R/R = 1.0/200 = 5 \times 10^{-3} \text{ A}$.

Voltmeter resistance: $R_V = V_V/I = 1.0/(5 \times 10^{-3}) = 200 \Omega$.

Final Answer

$$R_V = 200 \Omega; \quad I = 5 \times 10^{-3} \text{ A}$$

Question 51

The potential difference across a cell is 1.8 V when a current of 0.5 A is drawn from it. The PD falls to 1.6 V when a current of 1.0 A is drawn. Find the emf and the internal resistance of the cell.

Given: $V_1 = 1.8 \text{ V}$ at $I_1 = 0.5 \text{ A}$; $V_2 = 1.6 \text{ V}$ at $I_2 = 1.0 \text{ A}$

Formula: $V = \varepsilon - Ir$; two equations, two unknowns

Solution:

$$1.8 = \varepsilon - 0.5r \quad \dots (1)$$

$$1.6 = \varepsilon - 1.0r \quad \dots (2)$$

Subtracting (2) from (1):

$$0.2 = 0.5r \implies r = 0.4 \Omega$$

From (1): $\varepsilon = 1.8 + 0.5 \times 0.4 = 1.8 + 0.2 = 2.0 \text{ V}$

Final Answer

$$\varepsilon = 2.0 \text{ V}; \quad r = 0.4 \Omega$$

Question 52

A high-resistance voltmeter connected across the terminals of a battery reads 15 V. When an ammeter is included in the circuit, the voltmeter reads 9.0 V and the ammeter reads 1.5 A. Find (i) internal resistance of the battery and (ii) resistance of ammeter plus connecting wires.

Given: $\varepsilon = 15 \text{ V}$ (open circuit); $V_T = 9.0 \text{ V}$; $I = 1.5 \text{ A}$

Formula: $r = (\varepsilon - V_T)/I$; $R_{\text{ext}} = V_T/I$

Solution:

(i)

$$r = \frac{\varepsilon - V_T}{I} = \frac{15 - 9}{1.5} = \frac{6}{1.5} = 4.0 \Omega$$

(ii) Total external resistance:

$$R_{\text{total}} = \frac{\varepsilon}{I} - r = \frac{15}{1.5} - 4 = 10 - 4 = 6 \, \Omega$$

$$R_{\text{ammeter+wires}} = R_{\text{total}} - R_{\text{circuit}}$$

Since the voltmeter reads 9 V across the terminals and the ammeter reads 1.5 A, the external circuit (ammeter + wires) accounts for:

$$R_{\text{circuit}} = \frac{\varepsilon - V_T - 0}{I}$$

The total drop across external = 9 V for the voltmeter (terminals), so ammeter+wires voltage: $\varepsilon - V_T = 15 - 9 = 6$ V, and $R_{a+w} = 6/1.5 = 4 \, \Omega$...

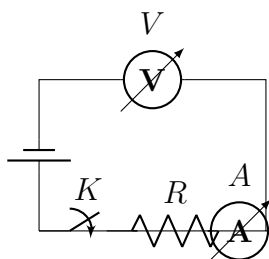
Reconsidering: the voltmeter reads terminal voltage = 9 V. The external resistance seen is $R_{\text{load}} = V_T/I = 9/1.5 = 6 \, \Omega$. This includes ammeter + wires. So $R_{\text{ammeter+wires}} = 6 \, \Omega$.

Final Answer

(i) $r = 4.0 \, \Omega$; (ii) $R_{\text{ammeter+wires}} = 6.0 \, \Omega$

Question 53

In the circuit shown, the resistance of the ammeter A is negligible and that of the voltmeter V is very high. When the key K is open, the reading of the voltmeter is 1.53 V. On closing the key, the ammeter reads 1.00 A and the voltmeter drops to 1.03 V. Calculate: (i) emf of the cell, (ii) internal resistance of the cell, (iii) value of R .



Given: $V_{\text{open}} = 1.53 \, \text{V}$; $I = 1.00 \, \text{A}$; $V_{\text{closed}} = 1.03 \, \text{V}$

Solution:

(i) **emf:**

$$\varepsilon = V_{\text{open}} = 1.53 \, \text{V}$$

(Voltmeter reads emf when key is open and voltmeter resistance is very high.)

(ii) **Internal resistance:**

$$r = \frac{\varepsilon - V_{\text{closed}}}{I} = \frac{1.53 - 1.03}{1.00} = 0.50 \, \Omega$$

(iii) **Resistance R :**

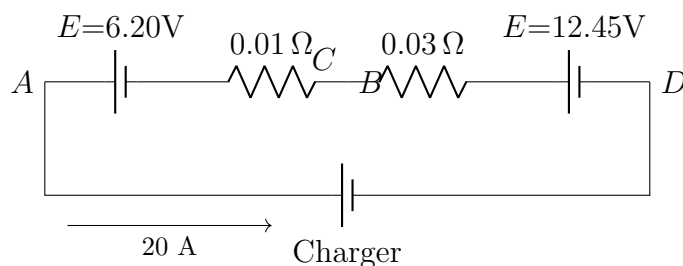
$$R = \frac{V_{\text{closed}}}{I} = \frac{1.03}{1.00} = 1.03 \, \Omega$$

Final Answer

(i) $\varepsilon = 1.53 \text{ V}$; (ii) $r = 0.50 \Omega$; (iii) $R = 1.03 \Omega$

Question 54

Two batteries of emfs 6.20 V and 12.45 V and internal resistances 0.01Ω and 0.03Ω respectively are being charged by a battery charger which supplies a current of 20 A . What are the terminal potential differences of the batteries during charging?



Given: Battery 1: $\varepsilon_1 = 6.20 \text{ V}$, $r_1 = 0.01 \Omega$; Battery 2: $\varepsilon_2 = 12.45 \text{ V}$, $r_2 = 0.03 \Omega$; $I = 20 \text{ A}$ (charging)

Formula: During charging: $V_T = \varepsilon + Ir$ (terminal PD rises above emf)

Solution:

Battery 1:

$$V_1 = \varepsilon_1 + Ir_1 = 6.20 + 20 \times 0.01 = 6.20 + 0.20 = 6.40 \text{ V}$$

Battery 2:

$$V_2 = \varepsilon_2 + Ir_2 = 12.45 + 20 \times 0.03 = 12.45 + 0.60 = 13.05 \text{ V}$$

Final Answer

$V_1 = 6.40 \text{ V}$; $V_2 = 13.05 \text{ V}$

Common Mistake: During charging, current flows into the positive terminal, so terminal PD $= \varepsilon + Ir$, not $\varepsilon - Ir$.

IX. Combination of Cells

Question 55

Three identical cells, each of emf 2 V and unknown internal resistance, are connected in parallel. This combination is connected to a $6\ \Omega$ resistor. The terminal potential difference across each cell is 1.5 V. What is the internal resistance of each cell?

Given: $n = 3$ identical cells in parallel; $\varepsilon = 2\text{ V}$ each; $R = 6\ \Omega$; $V_T = 1.5\text{ V}$

Formula: For n identical cells in parallel: effective emf $= \varepsilon$, effective internal resistance $= r/n$; $V_T = \varepsilon - I_{\text{total}} \cdot (r/n)$; $I_{\text{total}} = V_T/R$

Solution:

$$I = \frac{V_T}{R} = \frac{1.5}{6} = 0.25\text{ A}$$

$$V_T = \varepsilon - I \cdot \frac{r}{n} \implies 1.5 = 2 - 0.25 \times \frac{r}{3}$$

$$0.25 \times \frac{r}{3} = 0.5 \implies r = \frac{0.5 \times 3}{0.25} = 6\ \Omega$$

Final Answer

$$r = 6\ \Omega$$

Question 56

- (i) Three cells of emfs 2.0 V, 1.8 V and 1.5 V and internal resistances $0.05\ \Omega$, $0.7\ \Omega$ and $1.0\ \Omega$ respectively are joined in series. This battery is connected to an external resistor of $4.0\ \Omega$ through a low-resistance ammeter. What would the ammeter read?
- (ii) If these cells are joined in parallel, can the battery so formed give a definite emf and internal resistance?

Given: (i) $\varepsilon_1 = 2.0\text{ V}$, $r_1 = 0.05\ \Omega$; $\varepsilon_2 = 1.8\text{ V}$, $r_2 = 0.7\ \Omega$; $\varepsilon_3 = 1.5\text{ V}$, $r_3 = 1.0\ \Omega$; $R = 4.0\ \Omega$

Formula: Series cells: $\varepsilon_{\text{total}} = \sum \varepsilon_i$; $r_{\text{total}} = \sum r_i$; $I = \varepsilon_{\text{total}}/(R + r_{\text{total}})$

Solution:

(i)

$$\varepsilon_{\text{total}} = 2.0 + 1.8 + 1.5 = 5.3\text{ V}$$

$$r_{\text{total}} = 0.05 + 0.7 + 1.0 = 1.75\ \Omega$$

$$I = \frac{\varepsilon_{\text{total}}}{R + r_{\text{total}}} = \frac{5.3}{4.0 + 1.75} = \frac{5.3}{5.75} \approx 0.92\text{ A}$$

(ii) For non-similar cells in parallel, Kirchhoff's rules must be applied to find currents in different branches. A simple definite emf and internal resistance cannot be stated for non-identical cells in parallel.

Final Answer

(i) $I \approx 0.92$ A; (ii) No — for non-identical cells in parallel, Kirchhoff's laws must be used.

Question 57

Two cells, each of emf 1.5 V, are joined in series and send 1.0 A current through an external resistor. If the same cells are joined in parallel, then they send 0.6 A current through the same resistor. What is the internal resistance of each cell?

Given: Series: $\varepsilon_s = 2\varepsilon$, $r_s = 2r$, $I_s = 1.0$ A; Parallel: $\varepsilon_p = \varepsilon$, $r_p = r/2$, $I_p = 0.6$ A; $\varepsilon = 1.5$ V

Formula: Series: $I_s = 2\varepsilon/(R + 2r)$; Parallel: $I_p = \varepsilon/(R + r/2)$

Solution:

From series:

$$1.0 = \frac{2 \times 1.5}{R + 2r} = \frac{3}{R + 2r} \implies R + 2r = 3 \quad \dots (1)$$

From parallel:

$$0.6 = \frac{1.5}{R + r/2} \implies R + r/2 = \frac{1.5}{0.6} = 2.5 \quad \dots (2)$$

Subtracting (2) from (1):

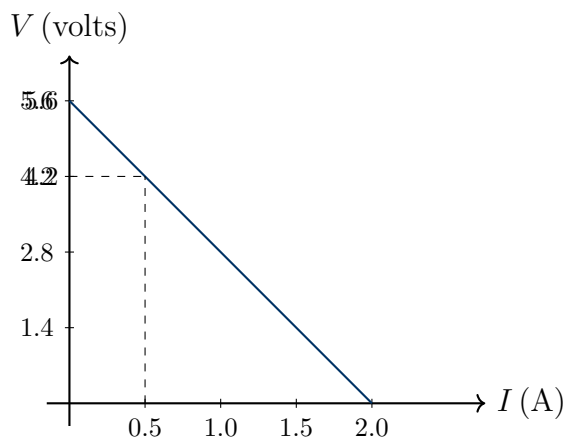
$$2r - r/2 = 3 - 2.5 = 0.5 \implies \frac{3r}{2} = 0.5 \implies r = \frac{1}{3} \Omega$$

Final Answer

$$r = \frac{1}{3} \Omega$$

Question 58

Four cells of identical emf E , internal resistance r , are connected in series to a variable resistor. The graph shows the variation of terminal voltage of the combination with current output. (i) What is the emf of each cell? (ii) For what current from the cells does maximum power dissipation occur? (iii) Calculate the internal resistance of each cell. (From graph: V -intercept = 5.6 V at $I = 0$; line passes through (0.5, 4.2) and has slope.)



Terminal voltage V versus current I for the series combination of 4 cells

Given: 4 cells in series; From graph: $V_0 = 5.6$ V (y-intercept); slope: using $(0, 5.6)$ and $(2.0, 0) \rightarrow \text{slope} = -5.6/2.0 = -2.8$ V/A

Formula: $V = 4E - 4rI$; Max power when $R = r_{\text{total}} = 4r$, i.e. $I = 4E/(2 \times 4r) = E/(2r)$

Solution:

(i) From graph: $4E = 5.6$ V $\Rightarrow E = 1.4$ V

(ii) Maximum power dissipation in external circuit occurs when $R_{\text{ext}} = r_{\text{total}}$.

The slope of the V - I graph $= -4r = -2.8 \Rightarrow r = 0.7$ Ω .

At maximum power, $R = 4r$:

$$I_{\text{max}} = \frac{4E}{4r + 4r} = \frac{E}{2r} = \frac{1.4}{2 \times 0.7} = 1 \text{ A}$$

(iii) From slope: $4r = 2.8 \Rightarrow r = 0.7$ Ω

Final Answer

(i) $E = 1.4$ V; (ii) $I = 1$ A; (iii) $r = 0.7$ Ω

★ Likely Exam Question

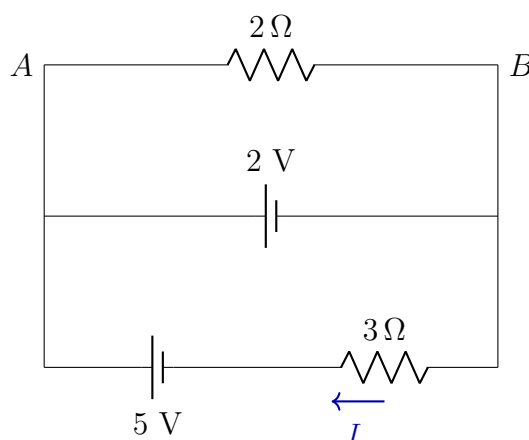
X. Kirchhoff's Laws

Question 59

Determine the current in the 3Ω resistance and the potential difference between A and B in the given circuit.

Given: Three parallel branches between nodes A and B : upper branch has 2Ω resistor; middle branch has 2 V cell (+ terminal at A); lower branch has 5 V cell (+ terminal at A) in series with 3Ω .

Solution:



Let $V_{AB} = V_A - V_B$. For each branch, write the KVL equation in the direction $A \rightarrow B$:

Middle branch (2 V cell, + at A , zero resistance): Going $A \rightarrow B$ we move from the + to the - terminal, so we lose 2 V :

$$V_A - 2 = V_B \quad \Rightarrow \quad V_{AB} = 2\text{ V}$$

Lower branch (5 V cell, + at A ; then 3Ω ; current I from A to B): Going $A \rightarrow B$:

$$V_A - 5 + 3I = V_B \quad \Rightarrow \quad V_{AB} = 5 - 3I$$

Setting both expressions for V_{AB} equal:

$$5 - 3I = 2 \quad \Rightarrow \quad 3I = 3 \quad \Rightarrow \quad I = 1\text{ A}$$

$$V_{AB} = 2\text{ V} \quad \Rightarrow \quad V_A - V_B = 2\text{ V}$$

Final Answer

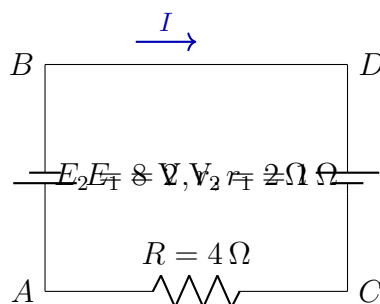
Current in 3Ω resistor = 1 A ; $V_A - V_B = 2\text{ V}$

Question 60

A cell of emf $E_1 = 2\text{ V}$ and internal resistance $r_1 = 1\ \Omega$ is connected to another cell of emf $E_2 = 8\text{ V}$ and internal resistance $r_2 = 2\ \Omega$ through an external resistance of $4\ \Omega$. Find the potential difference between point A and point C .

Given: $E_1 = 2\text{ V}$, $r_1 = 1\ \Omega$; $E_2 = 8\text{ V}$, $r_2 = 2\ \Omega$; $R = 4\ \Omega$

Solution:



Both cells are in a single loop. E_2 (8 V) opposes E_1 (2 V). Assume current I flows clockwise (driven by the larger E_2). Net emf driving current:

$$\varepsilon_{\text{net}} = E_2 - E_1 = 8 - 2 = 6\text{ V}$$

Total resistance:

$$R_{\text{total}} = r_1 + r_2 + R = 1 + 2 + 4 = 7\ \Omega$$

$$I = \frac{\varepsilon_{\text{net}}}{R_{\text{total}}} = \frac{6}{7}\text{ A (clockwise)}$$

Finding $V_A - V_C$: Traverse from A to C along the bottom wire (through $R = 4\ \Omega$). Current I flows from C to A through R (clockwise means current goes $A \rightarrow B \rightarrow D \rightarrow C$ inside the loop, so through R the current flows $C \rightarrow A$, i.e., from C up to A). Going from A to C , we go *against* the current through R :

$$V_A - V_C = -I \times R = -\frac{6}{7} \times 4 = -\frac{24}{7}\text{ V}$$

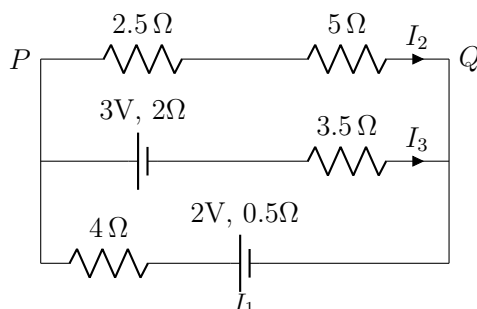
So $V_C - V_A = \frac{24}{7}\text{ V}$.

Final Answer

$$V_C - V_A = \frac{24}{7}\text{ V} \approx 3.43\text{ V}$$

Question 61

For the given electrical circuit, calculate the current through each branch using Kirchhoff's laws.



Given: Three parallel branches between nodes P and Q (as drawn):

Branch with current I_1 (bottom): 4Ω resistor in series with a 2 V cell of internal resistance 0.5Ω — total 4.5Ω .

Branch with current I_2 (top): 2.5Ω and 5Ω resistors in series, no cell — total 7.5Ω .

Branch with current I_3 (middle): a 3 V cell of internal resistance 2Ω in series with a 3.5Ω resistor — total 5.5Ω .

All three currents are taken as positive in the $P \rightarrow Q$ direction, and both cells have their $+$ terminal on the P side.

Solution:

Since all three branches connect the same two nodes, this is a two-node network — solve directly for $V = V_P - V_Q$, then get each branch current from it.

Branch equations (each written as current $= \frac{V - \varepsilon}{R}$, with $\varepsilon = 0$ where there is no cell):

$$I_1 = \frac{V - 2}{4.5}, \quad I_2 = \frac{V}{7.5}, \quad I_3 = \frac{V - 3}{5.5}$$

KCL at node P (current leaving P through the three branches must sum to zero, since P connects to nothing else):

$$I_1 + I_2 + I_3 = 0 \quad \Rightarrow \quad \frac{V - 2}{4.5} + \frac{V}{7.5} + \frac{V - 3}{5.5} = 0$$

Multiplying out and collecting terms:

$$399V = 735 \quad \Rightarrow \quad V = \frac{735}{399} = \frac{35}{19} \approx 1.84\text{ V}$$

Currents:

$$I_1 = \frac{35/19 - 2}{4.5} = -\frac{2}{57} \approx -0.04\text{ A}$$

$$I_2 = \frac{35/19}{7.5} = \frac{14}{57} \approx 0.25\text{ A}$$

$$I_3 = \frac{35/19 - 3}{5.5} = -\frac{4}{19} \approx -0.21\text{ A}$$

Check: $I_1 + I_2 + I_3 = -\frac{2}{57} + \frac{14}{57} - \frac{12}{57} = 0 \checkmark$

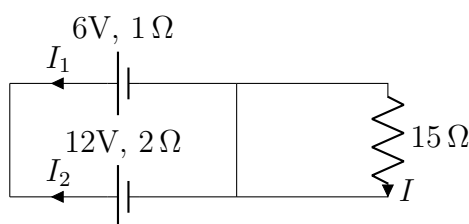
The negative signs on I_1 and I_3 mean those two branch currents actually flow from Q to P (opposite to the assumed direction).

Final Answer

$$I_1 \approx 0.04 \text{ A } (Q \rightarrow P); \quad I_2 \approx 0.25 \text{ A } (P \rightarrow Q); \quad I_3 \approx 0.21 \text{ A } (Q \rightarrow P)$$

Question 62

Two cells of emfs 6 V and 12 V and internal resistances 1Ω and 2Ω respectively are connected in parallel to send current in the same direction through an external resistance of 15Ω . Calculate current through each branch and PD across the 15Ω resistance.



Given: $\varepsilon_1 = 6 \text{ V}$, $r_1 = 1 \Omega$; $\varepsilon_2 = 12 \text{ V}$, $r_2 = 2 \Omega$; $R = 15 \Omega$

Solution:

Let I_1 = current through the 6 V cell, I_2 = current through the 12 V cell, $I = I_1 + I_2$ = current through R .

KVL — Loop 1 (6 V cell and R):

$$6 = I_1 \cdot 1 + 15I = I_1 + 15(I_1 + I_2) = 16I_1 + 15I_2 \quad (1)$$

KVL — Loop 2 (12 V cell and R):

$$12 = I_2 \cdot 2 + 15I = 2I_2 + 15(I_1 + I_2) = 15I_1 + 17I_2 \quad (2)$$

Multiply (1) by 17 and (2) by 15:

$$102 = 272I_1 + 255I_2$$

$$180 = 225I_1 + 255I_2$$

Subtract the first from the second:

$$78 = -47I_1 \Rightarrow I_1 = -\frac{78}{47} \text{ A}$$

The negative sign means the actual direction of I_1 is *opposite* to what we assumed — the 6 V cell is being charged by the 12 V cell. Substituting back into (1):

$$16\left(-\frac{78}{47}\right) + 15I_2 = 6 \Rightarrow 15I_2 = 6 + \frac{1248}{47} = \frac{282 + 1248}{47} = \frac{1530}{47} \Rightarrow I_2 = \frac{102}{47} \text{ A}$$

Current through R :

$$I = I_1 + I_2 = -\frac{78}{47} + \frac{102}{47} = \frac{24}{47} \text{ A}$$

Voltage across R :

$$V_{15} = IR = \frac{24}{47} \times 15 = \frac{360}{47} \approx 7.66 \text{ V}$$

Final Answer

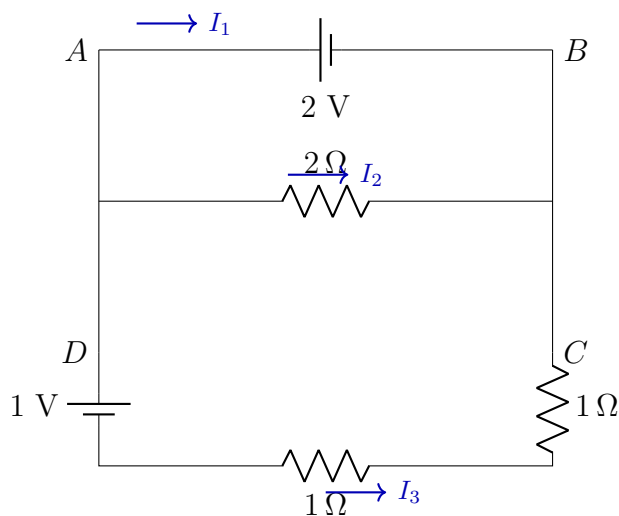
$|I_1| = \frac{78}{47} \text{ A}$ (through 6 V cell, reversed — cell being charged); $I_2 = \frac{102}{47} \text{ A}$ (through 12 V cell); $I = \frac{24}{47} \text{ A}$ through R ; $V_{15} \approx 7.66 \text{ V}$

Question 63

Calculate the currents I_1 , I_2 and I_3 in the given circuit. The cells have negligible internal resistances.

Given: 2 V cell directly across nodes A and B (branch 1, current I_1 , $A \rightarrow B$); 2Ω resistor across $A-B$ (branch 2, I_2); path $A \rightarrow D \rightarrow C \rightarrow B$ via a 1 V cell ($A-D$) and two 1Ω resistors (branch 3, I_3).

Solution:



The 2 V cell forces $V_A - V_B = 2 \text{ V}$ (taking $V_A > V_B$).

Branch 2 (2Ω , nodes $A-B$): Current from A to B through a resistor with 2 V across it:

$$I_2 = \frac{V_A - V_B}{2} = \frac{2}{2} = 1 \text{ A} \quad (A \rightarrow B)$$

Branch 3 (path $A \rightarrow D \rightarrow C \rightarrow B$): The 1 V cell raises potential from A to D by 1 V, so $V_D = V_A + 1 = 3 \text{ V}$ (taking $V_B = 0$, so $V_A = 2$). The two 1Ω resistors are in series from D to B with $(V_D - V_B) = 3 \text{ V}$ across them:

$$I_3 = \frac{V_D - V_B}{1 + 1} = \frac{3}{2} = 1.5 \text{ A} \quad (D \rightarrow C \rightarrow B)$$

This current flows $A \rightarrow D \rightarrow C \rightarrow B$ through branch 3, i.e., from A to B .

KCL at node A : Current leaving A = current entering A :

$$I_1(\text{into } A) = I_2(\text{leaving } A) + I_3(\text{leaving } A) = 1 + 1.5 = 2.5 \text{ A}$$

So $I_1 = 2.5 \text{ A}$ flows *into* node A through branch 1, meaning it flows from B to A inside the 2 V cell (the cell is being charged).

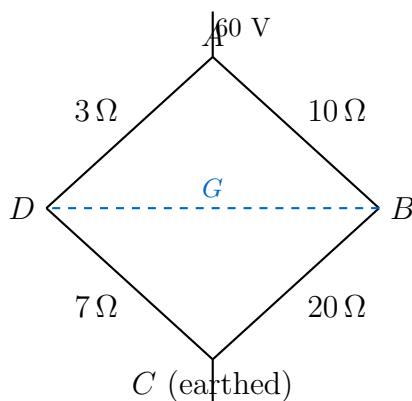
Final Answer

$I_1 = 2.5 \text{ A}$ (through 2 V cell, $B \rightarrow A$); $I_2 = 1 \text{ A}$ (through 2Ω , $A \rightarrow B$); $I_3 = 1.5 \text{ A}$ (through 1 V branch, $A \rightarrow D \rightarrow C \rightarrow B$)

XI. Wheatstone Bridge and Metre Bridge

Question 64

The given Wheatstone's bridge is almost balanced and the point C is earthed. (a) Find the potentials at points B and D . (b) If a galvanometer is connected across BD , what will be the direction of current in it? (c) For what value of the resistor DC would the bridge be balanced? Circuit: A at top; B on right; C at bottom (earthed); D on left. $AB = 10\ \Omega$, $BC = 20\ \Omega$, $AD = 3\ \Omega$, $DC = 7\ \Omega$; 60 V battery between A and C .



Wheatstone bridge with C earthed; battery of 60 V between A and C

Given: $R_{AB} = 10\ \Omega$, $R_{BC} = 20\ \Omega$, $R_{AD} = 3\ \Omega$, $R_{DC} = 7\ \Omega$; $V_C = 0\text{ V}$; $V_{AC} = 60\text{ V} \Rightarrow V_A = 60\text{ V}$

Solution:

(a) Potentials:

Branch $A \rightarrow B \rightarrow C$: Total $= 10 + 20 = 30\ \Omega$; current $= 60/30 = 2\text{ A}$.

$$V_B = V_C + I_{BC} \times 20 = 0 + 2 \times 20 = 40\text{ V}$$

Branch $A \rightarrow D \rightarrow C$: Total $= 3 + 7 = 10\ \Omega$; current $= 60/10 = 6\text{ A}$.

$$V_D = V_C + I_{DC} \times 7 = 0 + 6 \times 7 = 42\text{ V}$$

(b) Since $V_D = 42\text{ V} > V_B = 40\text{ V}$, current flows from D to B through the galvanometer.

(c) For balance: $\frac{R_{AB}}{R_{BC}} = \frac{R_{AD}}{R_{DC}}$

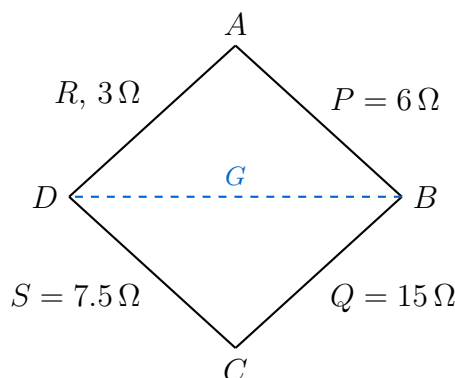
$$\frac{10}{20} = \frac{3}{R_{DC}} \Rightarrow R_{DC} = \frac{3 \times 20}{10} = 6\ \Omega$$

Final Answer

(a) $V_B = +40\text{ V}$, $V_D = +42\text{ V}$; (b) D to B ; (c) $R_{DC} = 6\ \Omega$

Question 65

The given circuit diagram represents a balanced Wheatstone's bridge. Calculate R . (Arms: $P = 6\ \Omega$; $Q = 15\ \Omega$; one arm contains R and $3\ \Omega$ in series; opposite arm contains $7.5\ \Omega$.)



Balanced Wheatstone bridge; galvanometer shows zero deflection

Given: Arm AB (ratio arm): $P = 6\ \Omega$; Arm BC : $Q = 15\ \Omega$; Arm AD : $R + 3\ \Omega$ (unknown); Arm DC (standard arm): $S = 7.5\ \Omega$

Formula: Balanced bridge: $\frac{P}{Q} = \frac{R_{AD}}{S}$

Solution:

For a balanced Wheatstone bridge, no current flows through the galvanometer, and the ratio of adjacent arms meeting at A equals the ratio of adjacent arms meeting at C :

$$\frac{P}{Q} = \frac{R_{AD}}{S} \Rightarrow \frac{6}{15} = \frac{R + 3}{7.5}$$

$$R + 3 = 7.5 \times \frac{6}{15} = 7.5 \times 0.4 = 3\ \Omega$$

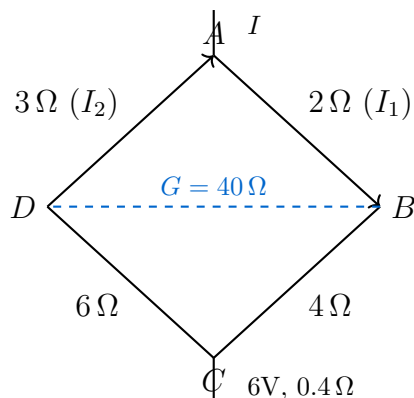
$$R = 3 - 3 = 0\ \Omega$$

Final Answer

$R = 0\ \Omega$ (Note: this follows directly from the arm values $P = 6\ \Omega$, $Q = 15\ \Omega$, $S = 7.5\ \Omega$ stated for this circuit. If your textbook prints a different final value, double-check the arm resistances against the original diagram, since a small transcription difference in any one of them would change R .)

Question 66

Calculate the currents I , I_1 and I_2 in the given Wheatstone's network of resistors. The internal resistance of the 6 V battery is $0.4\ \Omega$. (Network: $2\ \Omega$, $4\ \Omega$, $3\ \Omega$, $6\ \Omega$ bridge; galvanometer replaced with $G = 40\ \Omega$.)



Balanced Wheatstone network fed by a 6 V battery of internal resistance $0.4\ \Omega$

Given: $R_1 = 2\ \Omega$, $R_2 = 4\ \Omega$, $R_3 = 3\ \Omega$, $R_4 = 6\ \Omega$, $G = 40\ \Omega$; $\varepsilon = 6\ \text{V}$, $r = 0.4\ \Omega$

Solution:

First check if bridge is balanced: $R_1/R_2 = 2/4 = 0.5$; $R_3/R_4 = 3/6 = 0.5$. Since $R_1/R_2 = R_3/R_4$, the bridge is **balanced**, so no current through G .

Equivalent resistance (two series-parallel combinations):

$$R_{\text{top}} = R_1 + R_2 = 2 + 4 = 6\ \Omega; \quad R_{\text{bottom}} = R_3 + R_4 = 3 + 6 = 9\ \Omega$$

$$R_{\text{ext}} = \frac{6 \times 9}{6 + 9} = \frac{54}{15} = 3.6\ \Omega$$

$$I = \frac{\varepsilon}{R_{\text{ext}} + r} = \frac{6}{3.6 + 0.4} = \frac{6}{4.0} = 1.5\ \text{A}$$

$$I_1 = I \times \frac{R_{\text{bottom}}}{R_{\text{top}} + R_{\text{bottom}}} = 1.5 \times \frac{9}{15} = 0.9\ \text{A}$$

$$I_2 = I \times \frac{R_{\text{top}}}{R_{\text{top}} + R_{\text{bottom}}} = 1.5 \times \frac{6}{15} = 0.6\ \text{A}$$

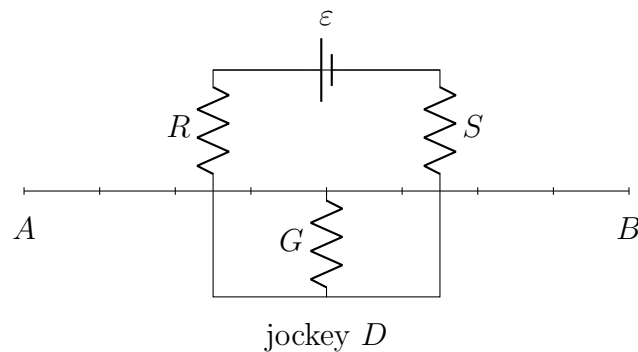
Final Answer

$$I = 1.5\ \text{A}; \quad I_1 = 0.9\ \text{A}; \quad I_2 = 0.6\ \text{A}$$

★ Likely Exam Question

Question 67

In a metre bridge, the null point is found at a distance of 60.0 cm from A. If now a resistance of $5\ \Omega$ is connected in series with S , the null point occurs at 50 cm. Determine the values of R and S .



Given: First null point: $l_1 = 60$ cm; Second null point (with $5\ \Omega$ in series with S): $l_2 = 50$ cm

Formula: Metre bridge: $\frac{R}{S} = \frac{l}{100 - l}$

Solution:

Case 1 (null at 60 cm):

$$\frac{R}{S} = \frac{60}{100 - 60} = \frac{60}{40} = \frac{3}{2} \Rightarrow R = \frac{3S}{2} \quad \dots (1)$$

Case 2 (null at 50 cm, $S' = S + 5\ \Omega$):

$$\frac{R}{S + 5} = \frac{50}{100 - 50} = \frac{50}{50} = 1 \Rightarrow R = S + 5 \quad \dots (2)$$

Substituting (1) into (2):

$$\frac{3S}{2} = S + 5 \Rightarrow \frac{3S - 2S}{2} = 5 \Rightarrow S = 10\ \Omega$$

$$R = \frac{3 \times 10}{2} = 15\ \Omega$$

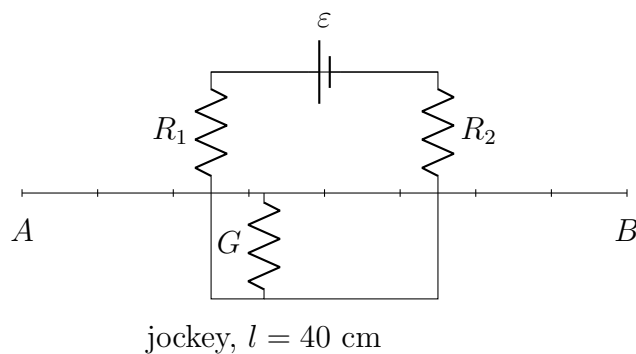
Final Answer

$$R = 15\ \Omega; \quad S = 10\ \Omega$$

★ Likely Exam Question

Question 68

In the experimental set-up of a metre bridge, the null point is obtained at 40 cm from A. If a $10\ \Omega$ resistor is connected in series with R_1 , the null point shifts by 10 cm. What resistance should be connected in parallel with $(R_1 + 10)\ \Omega$ such that the null point shifts back to its initial position?



Given: $l_1 = 40 \text{ cm}$; $l_2 = 50 \text{ cm}$ (after adding 10Ω to R_1); find R_p in parallel with $(R_1 + 10)$ to restore $l = 40 \text{ cm}$

Solution:

From initial balance:

$$\frac{R_1}{R_2} = \frac{40}{60} = \frac{2}{3} \implies R_1 = \frac{2R_2}{3} \quad \dots (1)$$

After adding 10Ω :

$$\frac{R_1 + 10}{R_2} = \frac{50}{50} = 1 \implies R_1 + 10 = R_2 \quad \dots (2)$$

$$\text{From (1) and (2): } \frac{2R_2}{3} + 10 = R_2 \implies 10 = R_2 - \frac{2R_2}{3} = \frac{R_2}{3} \implies R_2 = 30 \Omega$$

$R_1 = 20 \Omega$, so $R_1 + 10 = 30 \Omega$.

To restore null at 40 cm , need effective $R = R_1 = 20 \Omega$ in the first arm:

$$\frac{20}{30} = \frac{40}{60} \quad \checkmark$$

Let R_p be connected in parallel with 30Ω :

$$\frac{30 \times R_p}{30 + R_p} = 20 \implies 30R_p = 20(30 + R_p) = 600 + 20R_p$$

$$10R_p = 600 \implies R_p = 60 \Omega$$

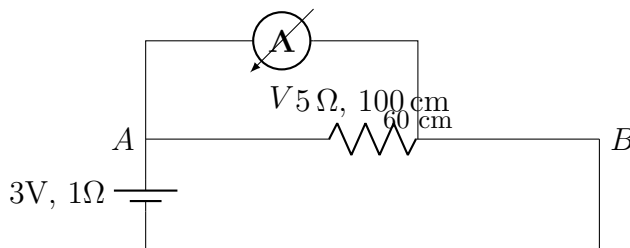
Final Answer

$$R_p = 60 \Omega$$

XII. Potentiometer

Question 69

A 3 V battery of internal resistance $1\ \Omega$ is joined to a sliding wire of length 100 cm and resistance $5\ \Omega$. A voltmeter, which takes negligible current, is connected across 60 cm length of the wire. (a) What is the reading of the voltmeter? (b) If the resistance of the voltmeter were $150\ \Omega$, what would have been its reading?



Given: $\varepsilon = 3\text{ V}$, $r = 1\ \Omega$; Wire: $L = 100\text{ cm}$, $R_w = 5\ \Omega$; $l = 60\text{ cm}$

Formula: Potential gradient $= \frac{\varepsilon}{r + R_w} \times \frac{R_w}{L}$; $V_{60} = \text{gradient} \times l$

Solution:

(a) **Voltmeter of negligible current (ideal):**

$$I = \frac{\varepsilon}{r + R_w} = \frac{3}{1 + 5} = \frac{3}{6} = 0.5\text{ A}$$

Potential gradient $= 0.5 \times 5/100 = 0.5 \times 0.05 = 0.025\text{ V/cm}$ (or $I \times R_{\text{per cm}} = 0.5 \times 5/100$).

$$V_{60} = 0.025 \times 60 = 1.5\text{ V}$$

(b) **Voltmeter resistance $R_V = 150\ \Omega$:**

Resistance of 60 cm wire $= 5 \times 60/100 = 3\ \Omega$.

Parallel combination of $3\ \Omega$ wire section and $150\ \Omega$ voltmeter:

$$R_{\text{par}} = \frac{3 \times 150}{3 + 150} = \frac{450}{153} = \frac{50}{17}\ \Omega$$

Remaining wire (40 cm) $= 5 \times 40/100 = 2\ \Omega$.

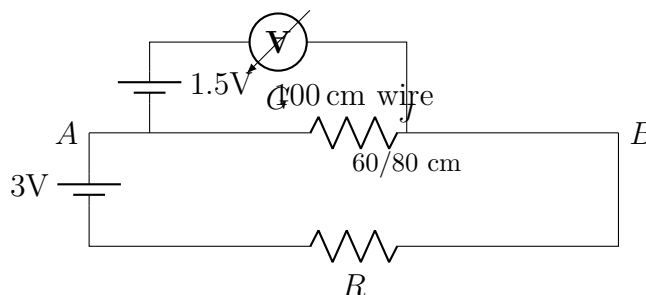
Total circuit resistance $= r + R_{40} + R_{\text{par}} = 1 + 2 + \frac{50}{17} = 3 + \frac{50}{17} = \frac{51+50}{17} = \frac{101}{17}\ \Omega$

$$I_{\text{total}} = \frac{3}{101/17} = \frac{3 \times 17}{101} = \frac{51}{101}\text{ A}$$

$$V_{\text{voltmeter}} = I_{\text{total}} \times R_{\text{par}} = \frac{51}{101} \times \frac{50}{17} = \frac{51 \times 50}{101 \times 17} = \frac{2550}{1717} = \frac{150}{101} \approx 1.485\text{ V}$$

Final Answer(a) $V = 1.5 \text{ V}$; (b) $V \approx 1.485 \text{ V}$ **Question 70**

A potentiometer wire of length 1 m is connected to a driver cell of emf 3 V. When a cell of emf 1.5 V is used in the secondary circuit, the balance point is found to be 60 cm. On replacing this cell with a cell of unknown emf, the balance point shifts to 80 cm. Calculate the unknown emf.



Given: $L_{\text{wire}} = 1 \text{ m}$; Driver emf = 3 V; $\varepsilon_1 = 1.5 \text{ V}$ at $l_1 = 60 \text{ cm}$; $\varepsilon_x = ?$ at $l_2 = 80 \text{ cm}$

Formula: $\varepsilon \propto l$; $\frac{\varepsilon_x}{\varepsilon_1} = \frac{l_2}{l_1}$

Solution:

$$\varepsilon_x = \varepsilon_1 \times \frac{l_2}{l_1} = 1.5 \times \frac{80}{60} = 1.5 \times \frac{4}{3} = 2.0 \text{ V}$$

Final AnswerUnknown emf $\varepsilon_x = 2.0 \text{ V}$ **Question 71**

A cell gives a balance with 85 cm of a potentiometer wire. When the terminals of the cell are shorted through a resistance of 7.5Ω , a balance is obtained at 75 cm. Find the internal resistance of the cell.

Given: $l_1 = 85 \text{ cm}$ (open circuit); $l_2 = 75 \text{ cm}$ (with $R = 7.5 \Omega$)

Formula: $r = R \left(\frac{l_1 - l_2}{l_2} \right)$

Solution:

$$r = R \times \frac{l_1 - l_2}{l_2} = 7.5 \times \frac{85 - 75}{75} = 7.5 \times \frac{10}{75} = 7.5 \times \frac{2}{15} = 1.0 \, \Omega$$

Final Answer

Internal resistance $r = 1.0 \, \Omega$

★ Likely Exam Question**Question 72**

A cell of emf 1.25 V gives a balance point at 35.0 cm length of a potentiometer wire. For another cell, the balance point shifts to 63.0 cm. What is the emf of the second cell?

Given: $\varepsilon_1 = 1.25 \, \text{V}$ at $l_1 = 35.0 \, \text{cm}$; $l_2 = 63.0 \, \text{cm}$

Formula: $\varepsilon \propto l$

Solution:

$$\varepsilon_2 = \varepsilon_1 \times \frac{l_2}{l_1} = 1.25 \times \frac{63.0}{35.0} = 1.25 \times 1.8 = 2.25 \, \text{V}$$

Final Answer

$\varepsilon_2 = 2.25 \, \text{V}$

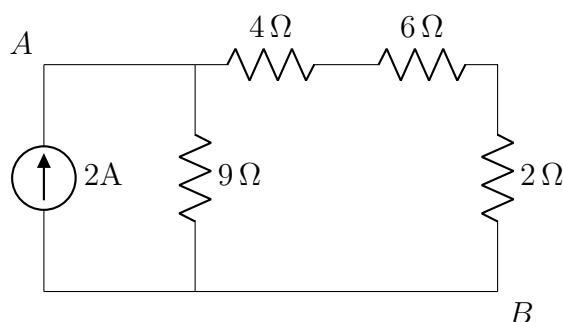
XIII. Miscellaneous

Question 73

Calculate in the given combination of resistors: (i) equivalent resistance between points A and B , (ii) PD between A and B , (iii) currents in the various resistors. A current of 2 A is supplied by the source.

Given: $4\ \Omega$ and $6\ \Omega$ in the top series branch ($A \rightarrow B$); $9\ \Omega$ in parallel across $A \rightarrow B$; $2\ \Omega$ at the far right; source current $I = 2\text{ A}$

Solution:



The $4\ \Omega$ and $6\ \Omega$ are in **series** (top path, A to B):

$$R_{\text{top}} = 4 + 6 = 10\ \Omega$$

The $2\ \Omega$ is in **series with** the $9\ \Omega$ branch? No — from the figure the $9\ \Omega$ is directly across $A-B$ (shunt) and the $2\ \Omega$ is also across $A-B$ in series with the $4\ \Omega$ – $6\ \Omega$ chain. So the three parallel paths between A and B are: $9\ \Omega$; $(4 + 6 + 2) = 12\ \Omega$.

(i) **Equivalent resistance:**

$$R_{AB} = \frac{9 \times 12}{9 + 12} = \frac{108}{21} = \frac{36}{7}\ \Omega$$

(ii) **PD between A and B :**

$$V_{AB} = I \times R_{AB} = 2 \times \frac{36}{7} = \frac{72}{7} \approx 10.3\text{ V}$$

(iii) **Currents:**

$$I_{9\Omega} = \frac{V_{AB}}{9} = \frac{72/7}{9} = \frac{72}{63} = \frac{8}{7}\text{ A}$$

$$I_{4\Omega} = I_{6\Omega} = I_{2\Omega} = \frac{V_{AB}}{12} = \frac{72/7}{12} = \frac{72}{84} = \frac{6}{7}\text{ A}$$

Check KCL: $I_{9\Omega} + I_{4\Omega} = \frac{8}{7} + \frac{6}{7} = \frac{14}{7} = 2\text{ A} \checkmark$

Final Answer

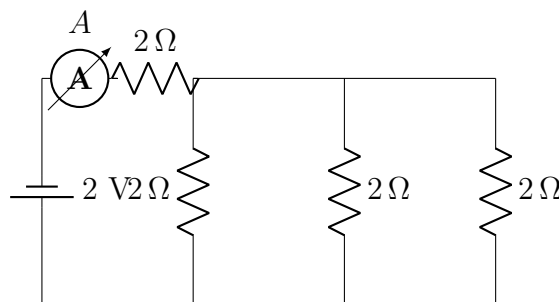
$$(i) R_{AB} = \frac{36}{7} \Omega \approx 5.14 \Omega; \quad (ii) V_{AB} = \frac{72}{7} \text{ V} \approx 10.3 \text{ V}; \quad (iii) I_{9\Omega} = \frac{8}{7} \text{ A}; \quad I_{4\Omega} = I_{6\Omega} = I_{2\Omega} = \frac{6}{7} \text{ A}$$

Question 74

Compute the reading of the ammeter A in the given network of resistors.

Given: 2 V cell (negligible internal resistance); 2Ω in series leading to junction point; three 2Ω resistors in parallel between the junction and the other terminal; ammeter A in the main series branch, between the cell and the junction, so it reads the total circuit current.

Solution:



The three 2Ω resistors are in **parallel**:

$$R_{\text{par}} = \frac{2}{3} \Omega$$

Total circuit resistance:

$$R_{\text{total}} = 2 + \frac{2}{3} = \frac{8}{3} \Omega$$

Total current from cell:

$$I_{\text{total}} = \frac{2}{8/3} = \frac{6}{8} = 0.75 \text{ A}$$

The ammeter sits in the main series branch, before the current splits into the three parallel resistors, so it reads the full circuit current:

$$I_A = I_{\text{total}} = 0.75 \text{ A}$$

Final Answer

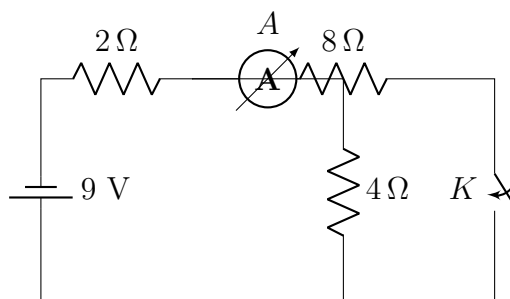
Ammeter reads $I_A = 0.75 \text{ A}$

Question 75

What will the ammeter A read when (a) the key K is off, (b) the key K is on? The resistance of the ammeter is negligible.

Given: 9 V source; 2Ω in series with the source; 4Ω resistor in one branch (with ammeter); key K in parallel with 8Ω resistor in the other branch.

Solution:



(a) Key K open:

With K open, the 8Ω branch is disconnected. Current flows only through the 4Ω branch (with ammeter). Total series resistance = $2 + 4 = 6\Omega$:

$$I = \frac{9}{2 + 4} = \frac{9}{6} = 1.5 \text{ A}$$

(b) Key K closed:

With K closed, the 8Ω resistor is bypassed (short-circuited), so the 8Ω branch now has 0Ω resistance. This 0Ω path is in **parallel** with the 4Ω (ammeter) branch:

$$R_{\text{par}} = \frac{4 \times 0}{4 + 0} = 0 \Omega$$

Total circuit resistance = $2 + 0 = 2\Omega$. Total current from source:

$$I_{\text{total}} = \frac{9}{2} = 4.5 \text{ A}$$

All current bypasses the 4Ω through the short circuit, so the **ammeter reads 0 A**. However, if K is in series with the 8Ω (not bypassing it), closing K connects the 8Ω in parallel with 4Ω :

$$R_{\text{par}} = \frac{4 \times 8}{4 + 8} = \frac{32}{12} = \frac{8}{3} \Omega; \quad I_{\text{total}} = \frac{9}{2 + 8/3} = \frac{9}{14/3} = \frac{27}{14} \text{ A}$$

$$I_A = I_{\text{total}} \times \frac{8}{4 + 8} = \frac{27}{14} \times \frac{8}{12} = \frac{27}{14} \times \frac{2}{3} = \frac{9}{7} \text{ A}$$

Final Answer

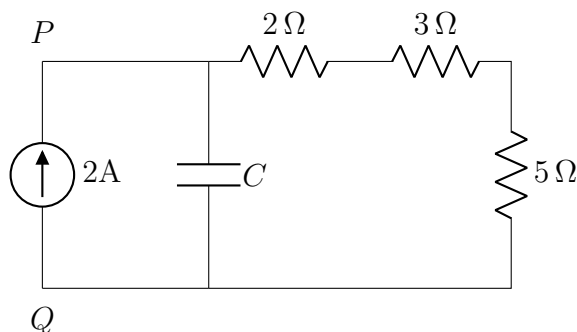
(a) K open: $I_A = 1.5 \text{ A}$; (b) K closed (8Ω added in parallel): $I_A = \frac{9}{7} \text{ A} \approx 1.29 \text{ A}$

Question 76

In steady state, calculate the energy stored in the capacitor included in the given circuit. ($I_{\text{source}} = 2 \text{ A}$, $C = 2 \mu\text{F}$.)

Given: Current source: $I = 2 \text{ A}$; 2Ω , 3Ω , 5Ω resistors; $C = 2 \mu\text{F}$ capacitor

Solution:



In **steady state**, a capacitor blocks DC — *no current flows through C* . All the source current (2 A) must therefore flow through the resistor chain $2 \Omega \rightarrow 3 \Omega \rightarrow 5 \Omega$ (total $= 10 \Omega$).

Voltage across the entire resistor chain (= voltage across the capacitor, since C is in parallel with the series chain $2 + 3 + 5 = 10 \Omega$):

$$V_C = I \times (2 + 3 + 5) = 2 \times 10 = 20 \text{ V}$$

Energy stored in the capacitor:

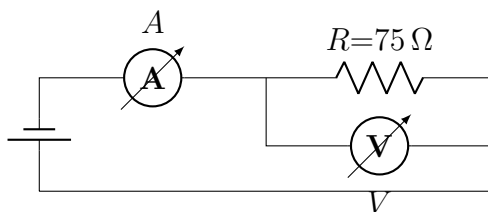
$$U = \frac{1}{2} C V_C^2 = \frac{1}{2} \times 2 \times 10^{-6} \times (20)^2 = \frac{1}{2} \times 2 \times 10^{-6} \times 400 = 4 \times 10^{-4} \text{ J}$$

Final Answer

$$U = 4 \times 10^{-4} \text{ J} = 0.4 \text{ mJ}$$

Question 77

In the given diagram, calculate the resistance of the voltmeter. The ammeter reading is 2 A , the voltmeter reading is 120 V and R is 75Ω .



Given: $I_A = 2 \text{ A}$; $V_V = 120 \text{ V}$; $R = 75 \Omega$

Solution:

The voltmeter (R_V) is connected in parallel with R . The ammeter reads the total current $I_A = 2 \text{ A}$, which splits between R and R_V .

Voltage across parallel combination = $V_V = 120 \text{ V}$.

Current through R : $I_R = V_V/R = 120/75 = 1.6 \text{ A}$.

Current through voltmeter: $I_V = I_A - I_R = 2 - 1.6 = 0.4 \text{ A}$.

$$R_V = \frac{V_V}{I_V} = \frac{120}{0.4} = 300 \Omega$$

Final Answer

$$R_V = 300 \Omega$$